F78AB

Actuarial and Financial Mathematics B



Department of Actuarial Mathematics and Statistics
Heriot-Watt University

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1 Discounted Cashflows

1.1 Yields Again

Consider the following cashflows representing (for example) profits from investment in a company or venture.



The yield on this transaction is 10% effective per time unit, because $i^* = 0.1$ is the solution of the equation of value:

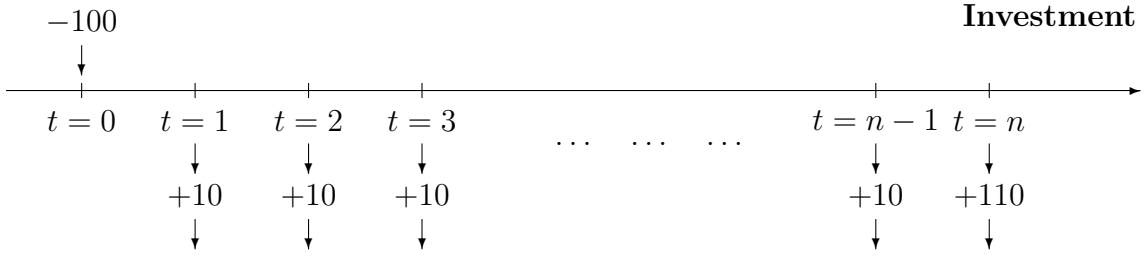
$$f(i) = -100(1+i)^n + 10s_{\overline{n}|}^{i} + 100 = 0.$$

We can rearrange this in a form recognizable as a statement of the Principle of Equivalence:

$$100 = 10\, a_{\overline{n}|}^{0.1} + 100\,(1.1)^{-n}.$$

The yield can be interpreted in terms of the Principle of Equivalence as follows: *if* money can be invested now and in the future (say in a bank account) to earn 10% per time unit effective interest, *then* we would be willing to exchange £100 now for the right to receive the future cashflows. The point here is that it really is necessary to be able to invest money to earn 10% per time unit — and this is not a given.

Looked at another way, if you invest in a project (or, indeed, buy a bond) that delivers cashflows to you at future points in time, you then have to *do something* with that money. Putting it in a bank account to earn more interest is an obvious option. In fact, it would be more realistic to split this investment into two parts: the part that delivers the cashflow to you, and the bank account in which you invest the cashflows:



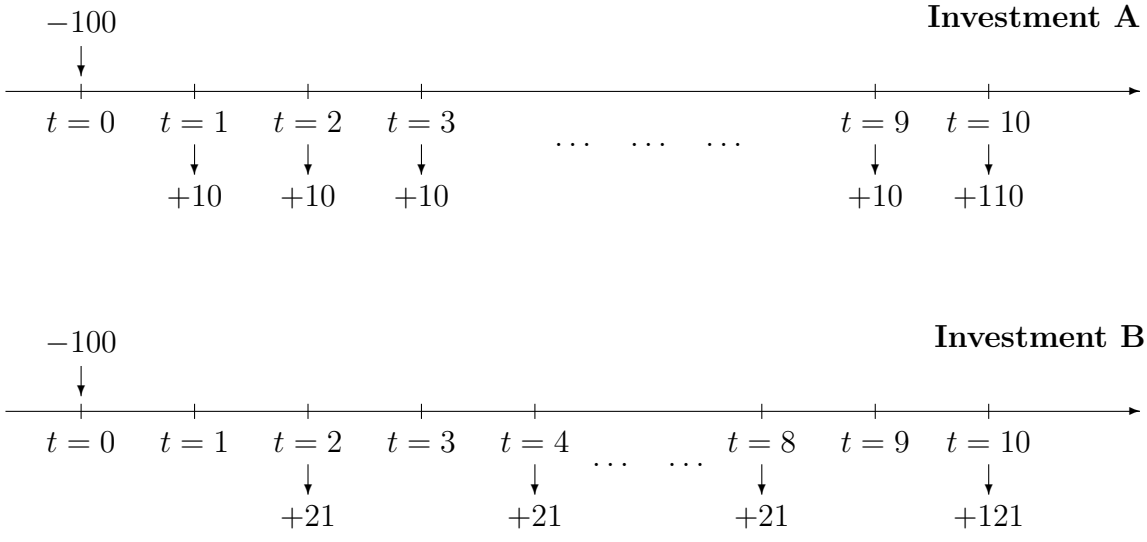


So the statement that the original investment has a yield of 10% per time unit effective is mathematically correct, but in the real world it only leads to an interpretation in terms of the Principle of Equivalence if market interest rates *actually are* 10% per time unit. If they are not, $i \neq 0.1$, then we see from the diagram above that once the intermediate cashflows are washed out, your transaction actually looks like the following:



And, it is the Principle of Equivalence we use to compare investments.

Consider the following two investments.



The yield on both investments is 10% per time unit effective (check this as an exercise). So *if* you could get a bank account that paid 10% interest per time unit effective, *then*:

- we would pay £100 for either investment; and

- we would be indifferent which of A or B we bought.

But, suppose you could get a bank account that paid 15% per time unit effective. Then note that:

$$\begin{aligned} 100\,(1.15)^{10} &= 404.56 \\ 100 + 10\,s_{\overline{10}|}^{0.15} &= 303.04 \\ 100 + 21\,s_{\overline{5}|}^{0.3225} &= 298.32. \end{aligned}$$

So now:

- we would not pay £100 for either investment; and
- we would rank A above B.

Questions: In view of the above, why are yields important? And, what exactly are 'market rates of interest'?

Answers: 'Market rates of interest' in fact *are* the yields observed in the bond markets, on which fixed interest securities of various terms are traded.

1.2 Net Present Values

Definition 1.1

Given a set of cashflows $c_1, c_2, \ldots, c_n$ at times $t_1, t_2, \ldots, t_n$, their *net present value function* (NPV) is defined as:

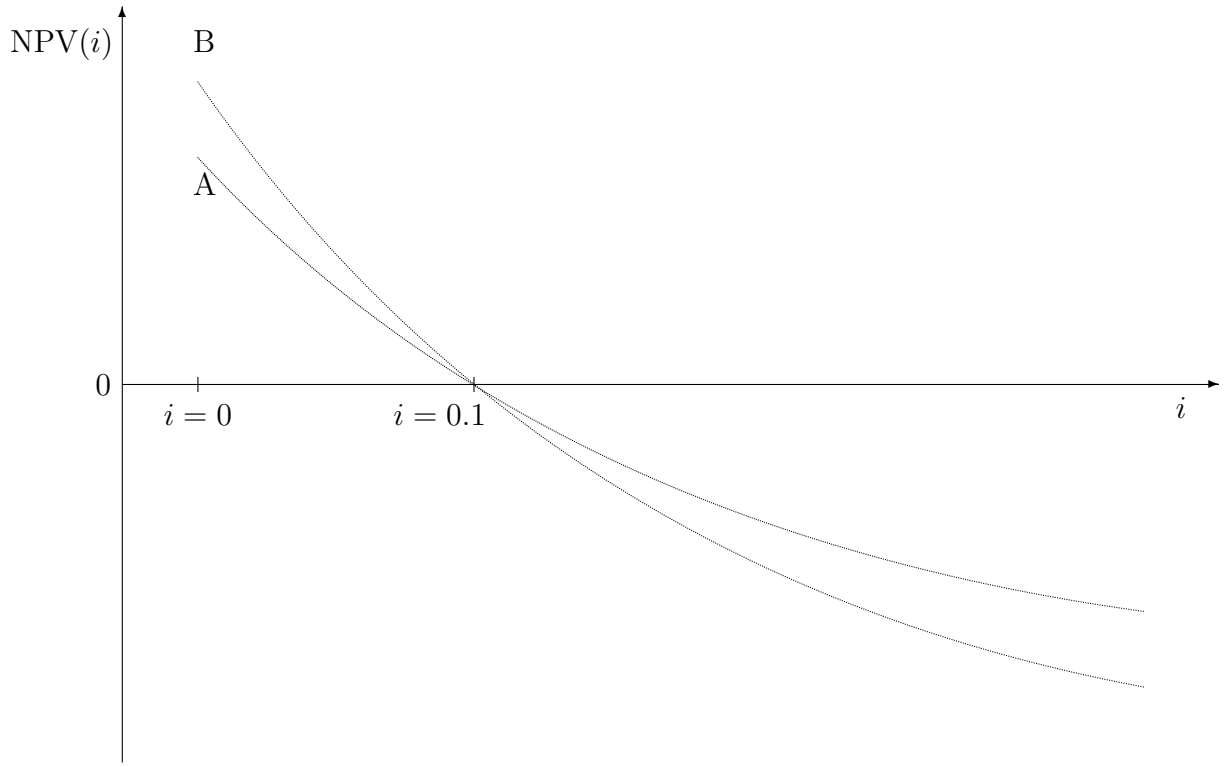
$$\text{NPV}(i) = \sum_{r=1}^{n} c_r\,(1+i)^{-t_r}.$$

That is, the present value of the cashflows, at time $t = 0$, as a function of the (constant) effective rate of interest i. By comparing NPVs of different investments at the market rate of interest we can choose between different investments (this is the Principle of Equivalence again).

Example 1.1 Consider Investment A and Investment B from the previous example. Their NPV functions are:

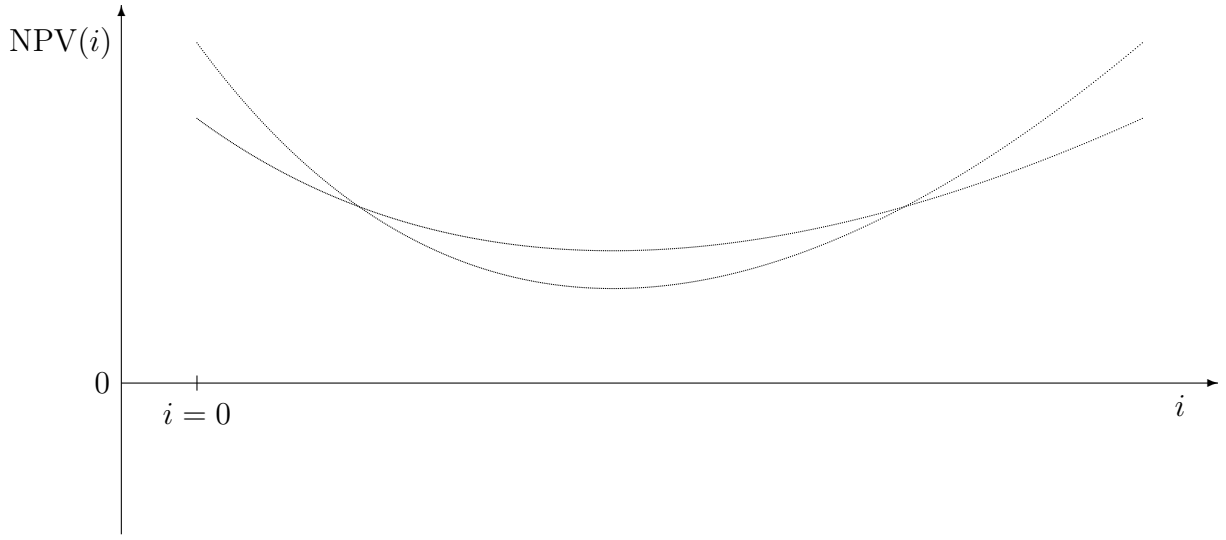
$$\begin{aligned} \text{NPV}_A(i) &= -100 + 10\,a_{\overline{10}|}^{i} + 100(1+i)^{-10} \\ \text{NPV}_B(i) &= -100 + 21\left((1+i)^{-2} + (1+i)^{-4} + (1+i)^{-6} + (1+i)^{-8} + (1+i)^{-10}\right) + 100(1+i)^{-10} \end{aligned}$$

We illustrate these functions in the following diagram.



Comments:

- The yield exists if and only if the equation NPV(i) = 0 has a unique solution not less than -1. However the NPV function always exists and is informative even if the yield does not exist. For example the diagram below shows the NPV functions of two investments that have no yield, but can still be meaningfully compared.



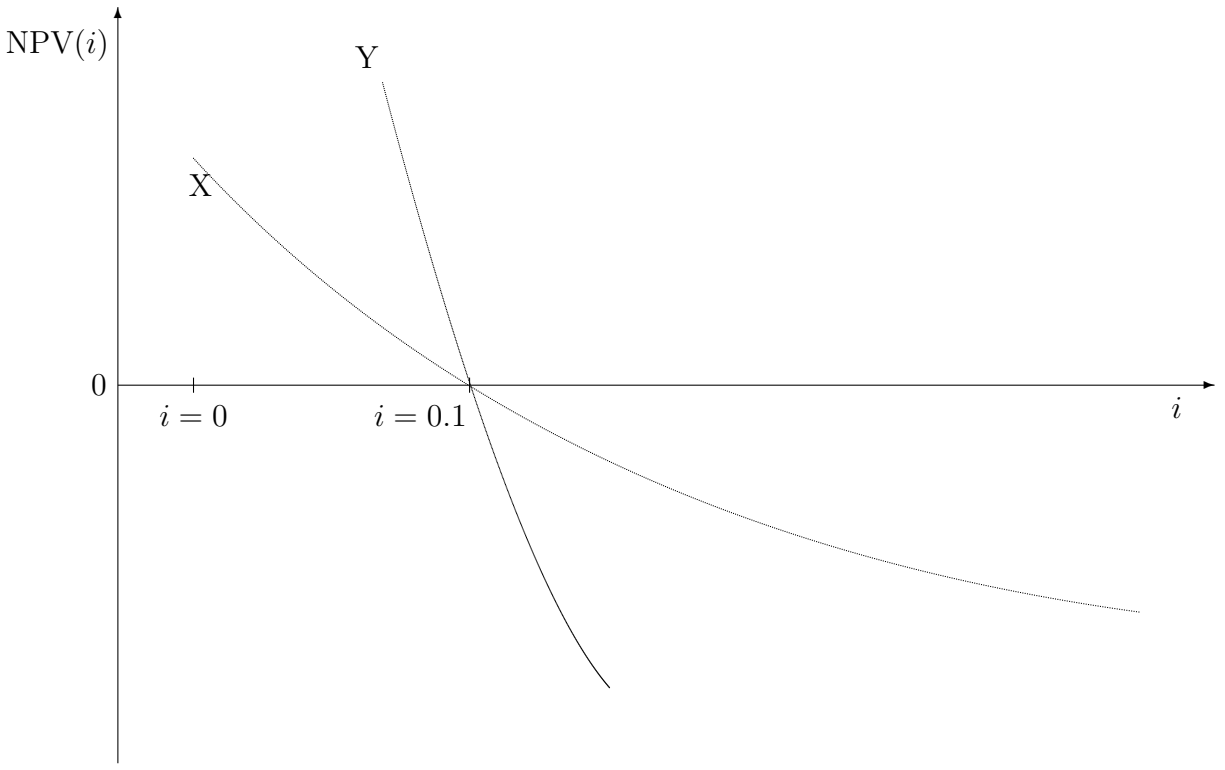
- The NPV functions of two investments may cross more than once, as in the previous examples. This means that we will prefer one to the other in different *regions* of market interest rates.

- NPV functions are additive: If we have two investments A and B and we combine their cashflows to create a new investment which we may call A+B, then:

$$\mathrm{NPV}_{A+B}(i) = \mathrm{NPV}_A(i) + \mathrm{NPV}_B(i).$$

  This is important when we form *investment portfolios* consisting of a collection of assets available in the market.

- Consider the two investments, X and Y, in the figure below.



  They have the same NPV at rate of interest i'. If the market interest rate is close to i', we should be indifferent between X and Y, according to the Principle of Equivalence. But Y is clearly much more sensitive to a *change* in the interest rate. For some purposes we would try to avoid this kind of sensitivity risk. Therefore, ranking of NPVs at a given interest rate (or the Principle of Equivalence) are not the only factors to be considered in practice.

- If the yield exists it is called the *internal rate of return* or IRR. This is the name most often used in corporate finance[1].

Example 1.2 Which of investments A or B in the previous example would you prefer if the market rate of interest was: (i) 5% p.a. effective?; (ii) 20% p.a. effective?

Answer: Calculate the NPVs of the two investments.

[1]The different name is a reflection of the fact that the concept of the yield has been reinvented several times, in different disciplines. We will meet yet another name for the yield in a later chapter.

(i) 5% p.a. effective.

$$\begin{aligned} \text{NPV}_A(0.05) &= 10\, a_{\overline{10}|}^{0.05} + 100\,(1.05)^{-10} - 100 = £38.61 \\ \text{NPV}_B(0.05) &= 21\, a_{\overline{5}|}^{0.1025} + 100\,(1.05)^{-10} - 100 = £ = 40.49. \end{aligned}$$

Hence we prefer Investment B.

(i) 20% p.a. effective.

$$\begin{aligned} \text{NPV}_A(0.20) &= 10\, a_{\overline{10}|}^{0.20} + 100\,(1.20)^{-10} - 100 = £ - 41.92 \\ \text{NPV}_B(0.20) &= 21\, a_{\overline{5}|}^{0.44} + 100\,(1.20)^{-10} - 100 = £ = -43.83. \end{aligned}$$

Hence we prefer Investment A, although we might also prefer to look elsewhere entirely.

1.3 The Discounted Payback Period

Most investments involve an initial outlay followed by income or profits. If an investment has a positive NPV (at interest rate i) there must be a first time when the initial outlay has been recovered, allowing for interest. This time is the *discounted payback period* (DPP), sometimes called the *break-even point*, at interest rate i .

Definition 1.2

Suppose we have a set of cashflows $c_1, c_2, \ldots, c_n$ at times $t_1, t_2, \ldots, t_n$, such that all negative cashflows precede all positive cashflows (investments followed by profits), and an effective rate of interest i per time unit such that $\text{NPV}(i) > 0$. The Discounted Payback Period (DPP) at effective rate of interest i per time unit is the smallest k such that:

$$\sum_{r=1}^{k} c_r (1+i)^{-t_r} \geq 0.$$

Example 1.3 You invest 100 at time $t = 0$, and receive casflows of 30 at times $t = 1, 2, 3, 4, 5$. The market interest rate of 5% p.a. effective. Find NPV(0.05) and the DPP.

Answer: $\text{NPV}(0.05) = -100 + 30\, a_{\overline{5}|}^{0.05} = 29.88$. This is positive, so the DPP exists and is the smallest k such that:

$$\begin{aligned} & 30\, a_{\overline{k}|}^{0.05} - 100 \geq 0 \\ \Rightarrow \quad & \frac{1-(1.05)^{-k}}{0.05} \geq \frac{10}{3} \end{aligned}$$

$$
\begin{aligned}
&\Rightarrow \quad (1.05)^{-k} \leq 0.83333\ldots \\
&\Rightarrow \quad (1.05)^{k} \geq 1.2 \\
&\Rightarrow \quad k \geq \frac{\log 1.2}{\log 1.05} \\
&\Rightarrow \quad k = 4 \text{ years.}
\end{aligned}
$$

Comment: You may think of the investment in terms of a loan schedule, as in Chapter **??**. The investment of 100 is treated as a 'loan' and each year's profit is applied first to repay 'interest' on the loan and then to repay a slice of the 'loan'. The DPP will automatically appear as the first repayment to reduce the capital content of the 'loan' to zero or less than zero. In the example above, the schedule would look like the following.

Time (Yrs)	'Repayment'	'Interest Content'	'Capital Content'	'Capital Outstanding'
0				100.00
1	30.00	5.00	25.00	75.00
2	30.00	3.75	26.25	48.75
3	30.00	2.44	27.56	21.19
4	30.00	1.06	28.94	−7.75
5	30.00	−0.39	30.39	−38.14

In the last year, the 'interest content' was negative because the 'loan account' was now in credit and earning interest rather then owing interest, and therefore the 'capital content' was larger than the 'repayment' itself. Note that $38.14\,(1.05)^{-5} = 29.88 = \text{NPV}(0.05)$.

1.4 Accumulated Profit

Complex investment decisions can be evaluated by working out the net cashflows and calculating the doscounted or accumulated profit. These can include money that is borrowed to finance projects, and the way in which borrowed money has to be repaid. We focus on accumulated profit.

Example 1.4 Suppose you can both borrow and invest money at 6% p.a. effective. You borrow £100 million, which is to be repaid by level annual repayments over ten years. The money finances a project that will make a profit of £18 million at the end of each of the *second* to tenth years inclusive.

(i) Find the net cashflows.

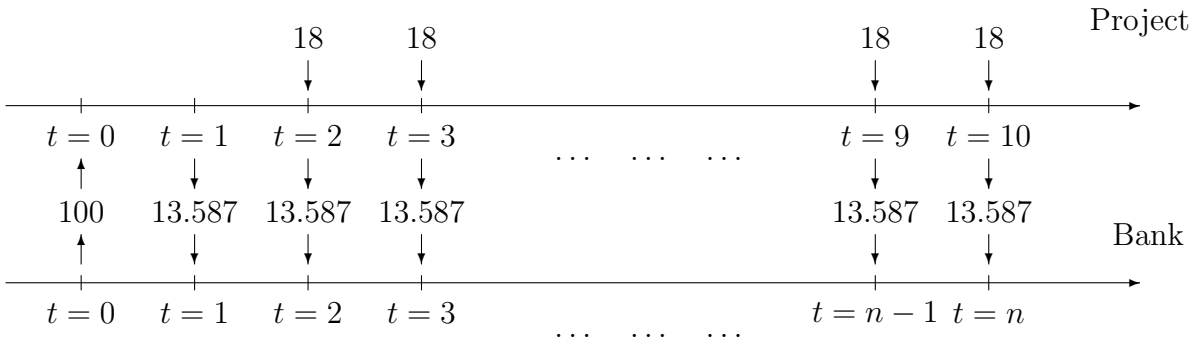
(ii) What is the accumulated profit at time $t = 10$?

(iii) What is NPV(0.06)?

For convenience, you may work in thousands of pounds.

Answers:

(i) The repayment is $100/a_{\overline{10}|}^{0.06}$ million = £13.587 million. The cashflows to and from the bank and the project are illustrated in the following related timelines



from which it is evident that the net cashflows to/from the project are those of the following timeline.



What happened to the initial £100 million? From the point of view of the borrower, it was money that came in (from the bank) and then went straight out again (spent on the project) so these cashflows cancelled out.

(ii) The accumulated profits are:

$$-13.587\,(1.06)^9 + 4.413\,s_{\overline{9}|}^{0.06} = £27.756 \text{ million.}$$

(iii) $\text{NPV}(0.06) = 27.756\,(1.06)^{-10} = £15.499$ million.

Methods that make direct use of cashflows such as these, allowing for several transactions at one between different parties, are known as Disounted Cashflow (DCF) methods. In the example above, the investor had to deal with cashflows to and from the project, and cashflows to and from the bank. There can be many parties involved in a complicated project.

DCF methods can be very useful because each cashflow is considered separately and incorporated into the investor's net position explicitly. This allows for factors that would be very difficult to

handle just using the traditional actuarial accumulated and discounted quantities. It does, also, mean that the use of a computer is necessary in any realistic example. We will illustrate its use to deal with one complicating factor, namely the fact that you cannot, in reality, borrow and invest money at the same rate of interest.

Example 1.5 You wish to borrow £100 to invest in a project that will return profits of £18 at the end of each of the next ten years. You borrow the money from a bank that charges interest of 8% p.a. effective. However, if you deposit or invest money you can earn only 6% p.a. effective interest. You have two possible ways to repay the loan,

(i) By net level repayments of £14.90 p.a., payable in arrears (check this!).

(ii) By using the entire £18 p.a. profit to repay the loan as soon as possible, thereafter investing the profits to earn 6% interest.

Which method would you prefer?

Comment: In this problem, we cannot easily work with discounted quantities, because we do not have a single, unambiguous rate of interest at which to do the discounting[2]. What we can calculate unambiguously is the accumulated profit after ten years.

Answer: As in the last example, the initial £100 is 'money in and money out' and disappears from the net cashflows. The investor has a net profit of (18.00 - 14.90) - £3.10 p.a., and can invest this to earn 6% p.a. effective interest, so the accumulated profit after ten years is $3.1\, s_{\overline{10}|}^{0.06} = £40.86$.

If the investor applies the entire profits to repay the loan as soon as possible we need to find the time when the loan is repaid, in other words, the DPP. This is the smallest time k such that the accumulated profits exceed the accumulated loan, that is such that:

$$\begin{aligned}
& 18\, s_{\overline{k}|}^{0.08} \geq 100\,(1.08)^k \\
\Rightarrow \quad & \frac{1-(1.08)^{-k}}{0.08} \geq \frac{100}{18} \\
\Rightarrow \quad & (1.08)^{-k} \leq 0.55555\ldots \\
\Rightarrow \quad & (1.08)^{k} \geq 1.8 \\
\Rightarrow \quad & k \geq \frac{\log 1.8}{\log 1.08} \\
\Rightarrow \quad & k = 8 \text{ years.}
\end{aligned}$$

It would be purely coincidental if the 8th profit paid off the loan exactly, so we need to calculate the outstanding loan at time $t = 8$. It is:

[2]Later, you will encounter a quantity called a *risk discount rate* which could be used to produce a present value in a case like this. It has its roots in the economic theory of risky investments and we will not deal with it here. We will say, though, that the risk discount rate would probably not be either 6% p.a. or 8% p.a. in this problem.

$$18\, s_{\overline{8}|}^{0.08} - 100\,(1.08)^8 = £6.37.$$

We can now accumulate this, and future profits, to time $t = 10$ at 6% p.a. effective:

$$\text{Accumulated profit} = 6.37\,(1.06)^2 + 18\,(1.06) + 18 = £44.24.$$

Compare this with £40.86 when the load was repaid by level repayments. Intuitively this makes sense; if we are having to pay a higher rate of interest on borrowings it is best to pay off the borrowed money as fast as possible.

If you compare the above to the 'loan schedule' shown in Section 1.3, you will see that we are doing just the same as we did there, except that when the loan was paid off the interest rate changed.

Example 1.6 An investor wishes to build a factory costing £100,000,000 to make computer games consoles. It is expected to produce profits of £50,000,000 at the end of each of the second, third, fourth and fifth years, after which it will be written off as valueless because the next generation of consoles will have arrived. The project can be financed in either of two ways as follows:

(A) The capital can be borrowed at 7% *per annum* effective interest. The profits will be used to pay off the loan as soon as possible, and once that is done, further profits can be invested to earn 5% p.a. effective interest.

(B) The investor can issue a bond with annual coupons of 6%, payable yearly in arrears, and term five years. It will be issued at par but must be redeemed at 110%. If the investor needs to borrow any more money it will be at 7% p.a. effective, while any profits can be invested to earn 5% p.a. effective.

Which method of funding should the investor choose? For convenience, you may work in thousands of pounds.

Answer: Since either method, potentially, involves more than one interest rate, we cannot compute a simple discounted quantity, but we can compute the accumulated profit after five years.

(A) We note that there is no profit at time $t = 1$, so by that time the loan will have accrued interest and will stand at £107 million. First we must find the DPP, when the loan is completely paid off. This is the smallest number of payments k such that:

$$\begin{aligned}
& 50\, s_{\overline{k}|}^{0.07} \geq 107\,(1.08)^k \\
\Rightarrow \quad & \frac{1-(1.07)^{-k}}{0.08} \geq \frac{107}{50} \\
\Rightarrow \quad & (1.07)^{-k} \leq 0.8502\ldots \\
\Rightarrow \quad & (1.07)^{k} \geq 1.1762 \\
\Rightarrow \quad & k \geq \frac{\log 1.1762}{\log 1.07}
\end{aligned}$$

$$\Rightarrow \quad k = 3 \text{ payments.}$$

The net profit to the investor when the loan is paid off is:

$$50\, s_{\overline{3}|}^{0.07} - 100\,(1.07)^4 = 29.665.$$

Hence the accumulated profit at time $t = 5$ is:

$$29.665\,(1.05) + 50 = £81.15 \text{ million.}$$

(B) The investor must borrow another £6 million at $t = 1$ to pay the coupon then due, as there is not yet any income. As in the last example, it is sensible to pay off as much as possible as soon as possible, so at time $t = 2$ the investor has net cashflow:

$$50 - 6 - 6\,(1.07) = £37.58 \text{ million.}$$

At $t = 3, 4, 5$ the investor's net cashflow is (50 - 6) = £44 million. Hence the accumulated profit at time $t = 5$ is:

$$37.58\,(1.05)^3 + 44\, s_{\overline{3}|}^{0.05} - 110 = £72.21 \text{ million.}$$

Hence the investor chooses method A.

2 Measuring Fund Performance

2.1 Fund Performance

Insurance companies and pension funds hold large funds on behalf of their members. Members pay premiums or contributions into the fund, which are invested by the fund manager on their behalf, and later their benefits or pensions are paid out of the fund. So money is constantly being paid into and out of the fund.

Clearly the members expect their fund manager to obtain as good a rate of return as possible. They want to measure this by comparing the rate of return on their fund with those achieved by other forms of investment, and by other fund managers.

In this chapter, we will look at how we can measure the performance of a fund manager, and consequently, how we can compare a fund with other investments.

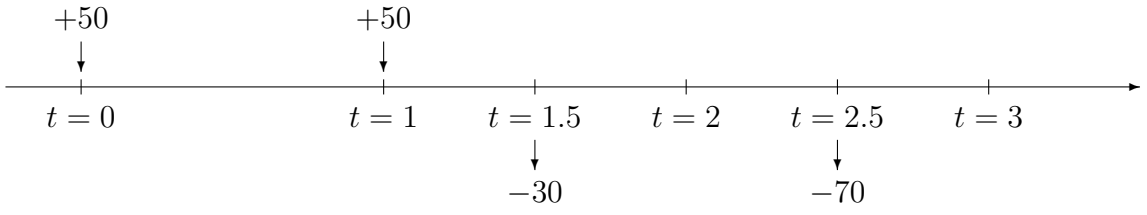
The salient features are the following.

- Although new funds do start up and old funds are wound up, a fund may be an ongoing entity. We may want to measure its performance over some period of time before which it existed and after which it will continue.
- Cashflows are coming into and out of the fund very frequently. A fund may have thousands or even millions of members, generating cashflows both in and out on a weekly, daily or even more frequent basis.

We have already considered the yield as a method of comparing alternative investments. We begin by considering it as a candidate measure of fund performance.

2.2 Yields Again

Example 2.1 Suppose that at time $t = 0$ a fund holds investments valued at 50 and the the cashflows in and out of the fund during the next three time units are as in the timeline below



and that at time $t = 3$ the fund holds investments valued at 80.

We can formulate an equation of value, choosing to do so at $t = 3$. In doing so, we treat the 50 already in the fund at time $t = 0$ as if it were a cashflow in at that time, because it certainly must

be included. Likewise, we treat the 80 in the fund at time $t = 3$ as if it were a cashflow out at that time.

$$f(i) = 100\,(1+i)^3 + 50\,(1+i)^2 - 30\,(1+i)^{1.5} - 70\,(1+i)^{0.5} - 80 = 0.$$

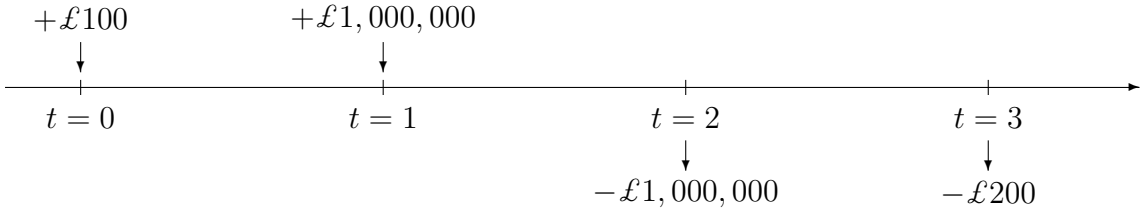
We can solve this for the yield, $i^* = 0.0856$ per time unit.

Comments:

- We set up a 'toy' example above, just to illustrate the ideas (and we will use it throughout this chapter). It has only four cashflows, whereas a real fund may have many thousands. Solving an equation of value in these circumstances may be very challenging and certainly messy.
- We contrived this example so that positive cashflows preceded negative cashflows, and by Steffensen's theorem the yield is sure to exist. With cashflows in and out of a real fund all the time, there is no guarantee that the yield will exist, and it might also be very difficult to try to check this.
- In the context of fund performance measurement, the yield is usually known as the *money-weighted rate of return* (MWRR), for reasons that will become apparent in the next example.

Example 2.2 Consider a fund with two members, A and B.

- At $t = 0$ it has nothing in it.
- Member A puts in £100 at time $t = 0$ and takes out £200 at time $t = 3$.
- Member B puts in £1 million at time $t = 1$ and takes out £1 million at time $t = 2$.
- at $t = 3$ the fund has nothing in it.



Investor A obtained a yield of $\left(\frac{200}{100}\right)^{1/3} - 1 \approx 0.26$ or about 26% per time unit and is presumably very happy. Investor B obtained a yield of nil and is presumably not so happy. If we formulate and solve the equation of value for the whole fund we write down:

$$f(i) = 100\,(1+i)^3 + 1,000,000\,(1+i)^2 - 1,000,000\,(1+i)^1 - 200 = 0$$

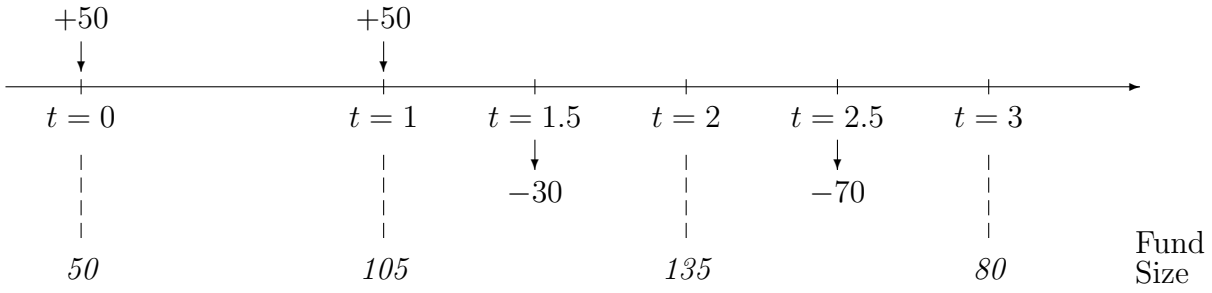
which we solve, getting $i^* \approx 0.0001$, or about 0.01% per time unit. The reason is that the equation of value is dominated by the largest cashflows. This is the origin of the name 'money-weighted' rate of return. It can be seen as a drawback. Here, clearly, the fund manager did very well indeed over the three time units as a whole, but has a poor performance in the middle of the period. Only the latter is revealed by the MWRR because the largest cashflows occurred in the middle of the time period.

2.3 Linked Internal Rates of Return

In order to calculate the yield, or MWRR, achieved on an ongoing fund, we had to know the value of the investments in the fund at the start and end of the period over which we wished to calculate the yield. So, in Example 1, the last cashflow was at time $t = 2.5$, but we did not know the fund size then, we knew the fund size at time $t = 0$ and $t = 3$ only. Therefore, we had to calculate the MWRR between those two times.

If we knew the fund size (the value of the investments held by the fund) at some intermediate times, we could calculate the yield over each of those sub-periods separately, and then link them together multiplicatively, using the corresponding accumulation factors, to find a yield over the whole period. This is called a *linked internal rate of return* (LIRR).

Example 2.3 Suppose in Example 2.1 we knew the fund size at some intermediate times, as follows. Note that the fund size at any time is given *before* any cashflow occurring at that time.



So, for example, the fund size is 105 at time $t = 1$ just before the cashflow of 50 into the fund, so at the start of the second sub-period the fund size will be $105 + 50 = 155$. These points where the fund size is known define three sub-periods, on each of which we can calculate a yield or MWRR. Define:

$$\begin{aligned} i_1 &= \text{the yield or MWRR on the period } t = 0 \text{ to } t = 1 \\ i_2 &= \text{the yield or MWRR on the period } t = 1 \text{ to } t = 2 \\ i_3 &= \text{the yield or MWRR on the period } t = 2 \text{ to } t = 3. \end{aligned}$$

Then on each sub-period, we formulate an equation of value, and solve it, as follows.

$$f(i_1) = 100\,(1+i_1)^1 - 105 = 0 \qquad \Rightarrow \qquad i_1^* = 0.05.$$

$$f(i_2) = 155\,(1+i_2)^1 - 30\,(1+i_2)^{0.5} - 135 = 0 \qquad \Rightarrow \qquad i_2^* = 0.0713.$$

$$f(i_3) = 135\,(1+i_3)^1 - 70\,(1+i_3)^{0.5} - 80 = 0 \qquad \Rightarrow \qquad i_3^* = 0.1482.$$

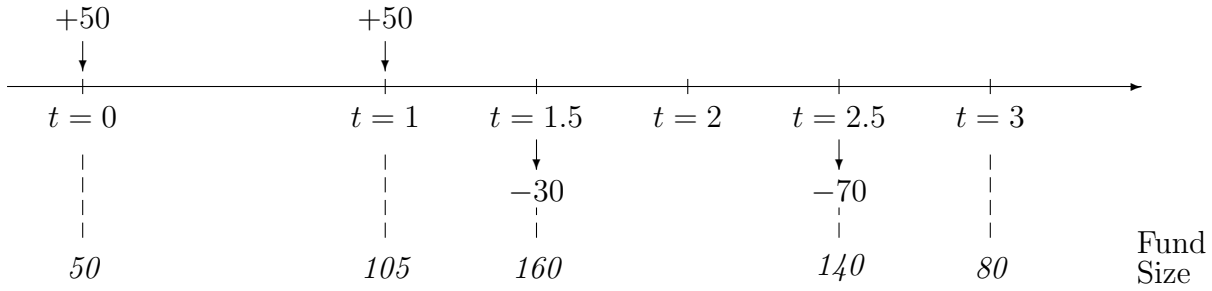
We then define the *linked internal rate of return* (LIRR) as the geometric average yield, weighted by the lengths of the sub-periods (which in this example are all the same).

$$1{+}\text{LIRR} = \left((1.05)^1(1.0713)^1(1.1482)^1\right)^{1/3} \qquad \Rightarrow \qquad \text{LIRR} = 0.089.$$

Hence, the LIRR is approximately 8.9% per time unit.

Comments:

- The LIRR is not uniquely defined. It depends on the sub-division of the whole period into sub-periods, and this is an essential part of the statement of a LIRR. For example, we may have chosen a different subdivision as follows.



we now have four subperiods, so define:

$$\begin{aligned} i_1 &= \text{the yield or MWRR on the period } t = 0 \text{ to } t = 1 \\ i_2 &= \text{the yield or MWRR on the period } t = 1 \text{ to } t = 1.5 \\ i_3 &= \text{the yield or MWRR on the period } t = 1.5 \text{ to } t = 2.5 \\ i_4 &= \text{the yield or MWRR on the period } t = 2.5 \text{ to } t = 3 \end{aligned}$$

and then formulate and solve an equation of value for the yield on each sub-period, as follows:

$$f(i_1) \;=\; 100\,(1+i_1)^1 - 105 = 0 \qquad \Rightarrow \qquad i_1^* = 0.05$$

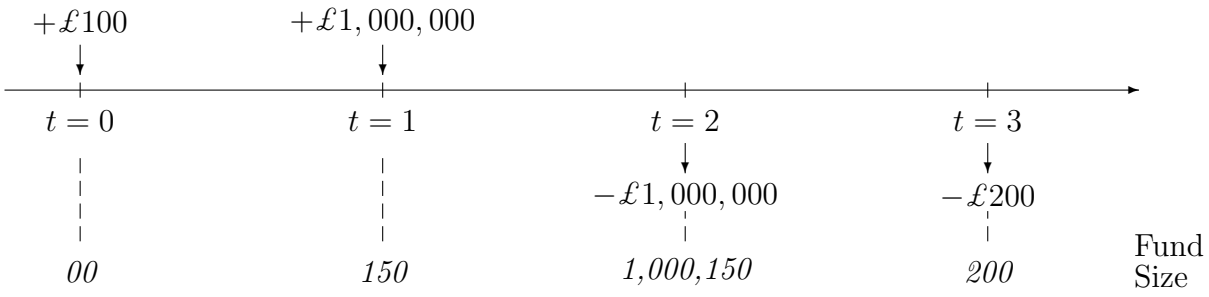
$$\begin{aligned} f(i_2) &= 155\,(1+i_2)^{0.5} - 160 = 0 \qquad \Rightarrow \qquad i_2^* = 0.0656 \\ f(i_3) &= 130\,(1+i_3)^{1} - 140 = 0 \qquad \Rightarrow \qquad i_3^* = 0.0769 \\ f(i_1) &= 70\,(1+i_4)^{0.5} - 80 = 0 \qquad \Rightarrow \qquad i_1^* = 0.3061. \end{aligned}$$

The LIRR) is then the geometric average yield, weighted by the lengths of the sub-periods (which in this example are not all the same).

$$1{+}\text{LIRR} = \left((1.05)^1(1.0656)^{0.5}(1.0769)^1(1.3061)^{0.5}\right)^{1/3} \qquad \Rightarrow \qquad \text{LIRR} = 0.1008.$$

Hence, the LIRR is approximately 10.1% per time unit. So the choice of sub-periods, in this case, made a difference of just over 1% to the result.

- It is clear, comparing the LIRRs above with the MWRR, that we gain much additional information from the process. We can now see, for example, that the fund performance was much influenced by an exceptionally good period in the last half time unit.
- The LIRR, with properly chosen sub-periods, is much less influenced by very large cashflows. Suppose in Example 2.2 we knew the fund sizes as follows.



So the relevant equations of value and yields on each sub-period are:

$$\begin{aligned} f(i_1) &= 100\,(1+i_1)^{1} - 150 = 0 \qquad \Rightarrow \qquad i_1^* = 0.50 \\ f(i_2) &= 1,000,150\,(1+i_2)^{0.5} - 1,000,150 = 0 \qquad \Rightarrow \qquad i_2^* = 0 \\ f(i_3) &= 150\,(1+i_3)^{1} - 200 = 0 \qquad \Rightarrow \qquad i_3^* = 0.333333. \end{aligned}$$

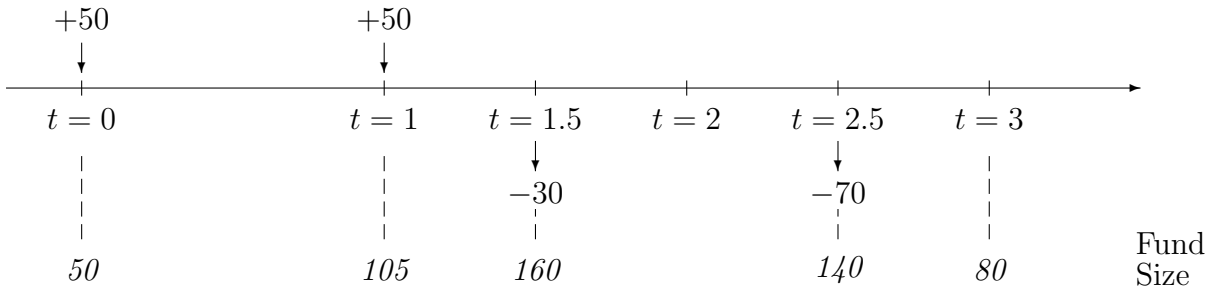
The LIRR) is then the geometric average yield, weighted by the lengths of the sub-periods:

$$1{+}\text{LIRR} = \left((1.50)^1(1.00)^1(1.333333)^1\right)^{1/3} \qquad \Rightarrow \qquad \text{LIRR} \approx 0.26.$$

Hence, the LIRR gives a more accurate impression of the fund performance over the whole period, eliminating the distortion cause by the large cashflows in the middle.

2.4 Time-Weighted Rate of Return

Suppose we know the fund size at the instant before every cashflow in or out of the fund (and at the beginning and the end, if there is not then a cashflow). We have actually already seen an example; it was the second LIRR calculation above based on Example 1.



Instead of calculating an LIRR, we can simplify the process, because there are no intervening cashflows between the times when the fund size is known. All we need do is calculate a *growth factor* for each period. Define:

$$\begin{aligned} g_1 &= \text{growth factor between } t=0 \text{ and } t=1 = \frac{105}{100} \\ g_2 &= \text{growth factor between } t=1 \text{ and } t=1.5 = \frac{160}{155} \\ g_3 &= \text{growth factor between } t=1.5 \text{ and } t=2.5 = \frac{140}{130} \\ g_4 &= \text{growth factor between } t=2.5 \text{ and } t=3 = \frac{80}{70}. \end{aligned}$$

The total growth factor over the whole period, as if we had invested 1 unit for the whole period, is just:

$$\frac{105}{100} \times \frac{160}{155} \times \frac{140}{130} \times \frac{80}{70} = 1.334.$$

Then the *time-weighted rate of return* (TWRR) is defined to be the geometric average rate of return per time unit, based on the total growth factor:

$$(1 + \text{TWRR}) = (\text{total growth factor})^{1/3} = 1.1008$$

so the TWRR is 10.08% per time unit.

Comment: The TWRR is identical to the LIRR if the times at which the fund size is known that define the sub-periods used to calculate the LIRR include, as a subset, the times at which all the cashflows occur.

3 Real interest rates and index-linked securities

3.1 The changing price of sugar

Suppose I buy 1kg of sugar every year on 1 January. How much does it cost me to buy 1kg of sugar? How does the cost change from year to year?

Suppose I buy 1kg of sugar at the start of every month. Suppose the prices I pay in pence each month are given in Table 1.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2000	166.6	167.5	168.4	170.1	170.7	171.1	170.5	170.5	171.7	171.6	172.1	172.2
2001	171.1	172.0	172.2	173.1	174.2	174.4	173.3	174.0	174.6	174.3	173.6	173.4
2002	173.3	173.8	174.5	175.7	176.2	176.2	175.9	176.4	177.6	177.9	178.2	178.5
2003	178.4	179.3	179.9	181.2	181.5	181.3	181.3	181.6	182.5	182.6	182.7	183.5
2004	183.1	183.8	184.6	185.7	186.5	186.8	186.8	187.4	188.1	188.6	189.0	189.9
2005	188.9	189.6	190.5	191.6	192.0	192.2	192.2	192.6	193.1	193.3	193.6	194.1

Table 1: The changing price in pence of 1kg of sugar.

Example 3.1 The change in the price of 1kg of sugar from January 2000 to January 2005 is

$$\frac{\text{Price of 1kg of sugar in January 2005}}{\text{Price of 1kg of sugar in January 2000}} = \frac{188.9}{166.6} = 1.13385,$$

i.e. the price of 1kg of sugar has increased by 13.385% over 5 years. Expressed as an annual effective rate, the price change is

$$\left(\frac{\text{Price of 1kg of sugar in January 2005}}{\text{Price of 1kg of sugar in January 2000}}\right)^{1/5} = 1.02544,$$

i.e. the annualised average change in the price of 1kg of sugar is +2.544% p.a. over 5 years.

Example 3.2 With £16.66 in January 2000, I can buy 10kg of sugar.

With £16.66 in December 2005, how many kg of sugar can I buy?

Solution: From Table 1, the cost of 1kg of sugar in December 2005 is £1.941. I can buy

$$\frac{16.66}{1.941} = 8.58\text{kg of sugar in December 2005.}$$

The *purchasing power* of £16.66 has fallen since 2000. It has fallen because the price of sugar has increased.

Who might care about the changes in the price of 1kg of sugar? More widely, who might care about the changes in the prices of goods and services?

In January 2000, I buy £100 nominal of a special 5-year bond, called a "Sugar Bond". The Sugar Bond pays no coupons but its redemption value is increased by the change in the price of sugar over the 5 years from January 2000 to January 2005.

Exercise 3.3 Using Table 1, calculate the redemption payment at January 2005.

Exercise 3.4 Calculate how much sugar can be bought with

1. £100 in January 2000.
2. The redemption payment in January 2005.

The *real value* of the redemption payment from the Sugar Bond stays constant. This is because the redemption payment changes proportionally to the change in the price of sugar.

The *money value* of the redemption payment £100 increases to £113.38535. The money value can be thought of as the number written on a banknote or a coin. Sometimes you may see the money value called the *nominal value* or the *monetary value*.

3.2 Price inflation indices

More generally, price inflation is the change of purchasing power of money. It is measured with reference to an index. Two important indices in the U.K. are

- Retail Prices Index (RPI),
- Consumer Prices Index (CPI).

They measure the monthly change in the prices of a specified set of goods and services. The Office for National Statistics (ONS) publish these indices.

Exercise 3.5 From the ONS website, find out how these two indices differ.

Exercise 3.6 Find out who makes use of RPI and CPI.

Example 3.7 You have started your dream job. At the end of your first year, your employer offers you a pay rise of 1%. You hear in the news that RPI has increased by 4% over the year and is expected to stay high in the next few years. Would you be happy with your pay rise?

Example 3.8 In 2012, you retired from your dream job. In 1962, you had deposited £100 000 in a savings account to provide for your retirement. The account has paid interest at an average effective rate of 6.14% per annum. Thus in 2012, the deposit was worth £1 967 679.

The average *money return* on the deposit is 6.14% per annum.

However, the RPI has increased at an average effective rate of 5.85% per annum from 1962 to 2012. Thus, in terms of 1962 purchasing power, the accumulated 2012 deposit has a real value of

$$\frac{1\,967\,679}{1.0585^{2012-1962}} = 114\,660.$$

In other words, £1 967 679 can buy as much in 2012 as £114 660 could buy in 1962.

The average *real return* or *real yield* on the deposit is +0.27% per annum, which can either be calculated from

$$\frac{1.0614}{1.0585} - 1 = 0.0027 \text{ per annum},$$

or from

$$\left(\frac{114\,660}{100\,000}\right)^{1/50} - 1 = 0.0027 \text{ per annum.}$$

Retail Prices Index (RPI)

- The RPI measures the monthly change in the prices of a specified set of goods and services.
- It is published one month in arrears (usually in the middle of the following month).

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2005	188.9	189.6	190.5	191.6	192.0	192.2	192.2	192.6	193.1	193.3	193.6	194.1
2006	193.4	194.2	195.0	196.5	197.7	198.5	198.5	199.2	200.1	200.4	201.1	202.7
2007	201.6	203.1	204.4	205.4	206.2	207.3	206.1	207.3	208.0	208.9	209.7	210.9
2008	209.8	211.4	212.1	214.0	215.1	216.8	216.5	217.2	218.4	217.7	216.0	212.9
2009	210.1	211.4	211.3	211.5	212.8	213.4	213.4	214.4	215.3	216.0	216.6	218.0
2010	217.9	219.2	220.7	222.8	223.6	224.1	223.6	224.5	225.3	225.8	226.8	228.4
2011	229.0	231.3	232.5	234.4	235.2	235.2	234.7	236.1	237.9	238.0	238.5	239.4
2012	238.0	239.9	240.8	242.5	242.4	241.8	242.1	243.0	244.2	245.6	245.6	246.8
2013	245.8	247.6	248.7	249.5	250.0	249.7	249.7	251.0	251.9	251.9	252.1	253.4
2014	252.6	254.2	254.8	255.7	255.9	256.3	256.0	257.0	257.6	257.7	257.1	257.5

Table 2: RPI since 2005. Earlier figures are identical to those in Table 1.

Example 3.9 The change in RPI from October 2012 to October 2013 is

$$\frac{\text{RPI(Oct 13)}}{\text{RPI(Oct 12)}} - 1 = \frac{251.9}{245.6} - 1 = 0.02565,$$

i.e. the prices of the basket of goods and services on which the RPI is based has increased by +2.565%.

Example 3.10 The change in RPI from October 2011 to October 2012 is

$$\frac{\text{RPI(Oct 12)}}{\text{RPI(Oct 11)}} - 1 = \frac{245.6}{238.0} - 1 = 0.03193,$$

i.e. the prices of the basket of goods and services on which the RPI is based has increased by +3.193%.

Example 3.11 The annualised average change in RPI from October 2011 to October 2013 is

$$\left(\frac{\text{RPI(Oct 13)}}{\text{RPI(Oct 11)}}\right)^{1/2} - 1 = \left(\frac{251.9}{238.0}\right)^{1/2} - 1 = 0.02879,$$

i.e. the prices of the basket of goods and services on which the RPI is based has increased at an average effective rate of +2.879% per annum.

Exercise 3.12 Calculate the annualised average change in RPI from January 2005 to November 2013.

Exercise 3.13 Does the RPI always increase?

Example 3.14 In the 2010 Budget, the Government announced that "The Government will adopt the CPI for the indexation of benefits, tax credits and public service pensions from April 2011" (Source: HM Treasury, *Budget 2010*, page 17, paragraph 1.43). Previously, RPI had been used.

- What reason did the Government give for the change?
- Is there an impact on the recipients of the benefits, tax credits and pensions?

3.3 Real yields

In F78AA, you calculated the *money yield* yield on assets, projects and bonds. The money yield uses the money payments.

We consider how to calculate the *real yield*. This is the yield which is calculated using the real value of payments. Although the real value of payment is relative to a chosen fixed date, which we call the *base date*, it turns out the real yield doesn't depend on which base date we use.

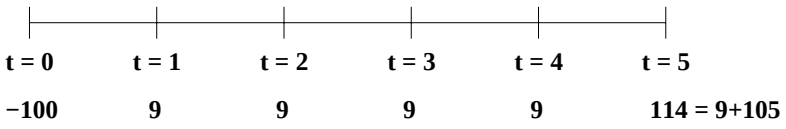
Here we use i to denote the money yield and use i' to denote the real yield.

Example 3.15 (Real yields vs. money yields)

Consider a 5-year fixed-interest bond issued in January 2005 with a coupon of 9% per annum paid annually and a redemption rate of 105%. The issue price is £100 per £100 nominal.

Question For an investor not subject to tax, calculate the yield:

1. In money terms.
2. In real terms relative to Jan 2005 with reference to the RPI.



Solution

1. Solving $100 = 9a_{\overline{5}|} + 105v^5$ for an effective interest rate i p.a, we find $i = 9.822\%$ p.a.
2. To adjust the bond payments to real values relative to January 2005, we divide each payment by the RPI at the payment date. Then we multiply by the RPI at January 2005. The calculations are shown below.

Time t	0	1	2	3	4	5
Money cashflow	-100	9	9	9	9	114
Payment date	Jan 2005	Jan 2006	Jan 2007	Jan 2008	Jan 2009	Jan 2010
RPI(Payment date)	188.9	193.4	201.6	209.8	210.1	217.9
Real cashflow relative to base date Jan 2005	$\frac{188.9}{188.9} \times -100$ $= -100$	$\frac{188.9}{193.4} \times 9$ $= 8.79059$	$\frac{188.9}{201.6} \times 9$ $= 8.43304$	$\frac{188.9}{209.8} \times 9$ $= 8.10343$	$\frac{188.9}{210.1} \times 9$ $= 8.09186$	$\frac{188.9}{217.9} \times 114$ $= 98.82790$

The real yield i' p.a. effective is the solution to:

$$100 = 8.79059v + 8.43304v^2 + 8.10343v^3 + 8.09186v^4 + 98.82790v^5 \quad \Rightarrow \quad i' = 6.705\%\text{p.a.}$$

Exercise 3.16 The base RPI date at which the real values are calculated (January 2005) is irrelevant for the calculation of i'. By considering the equation of value, explain why.

Exercise 3.17 What is the relationship between the money yield and the real yield?

Example 3.18 Today you deposit £1 000 in a bank account paying interest at 0% per annum. What will be the real value of the £1 000 in (i) one year's time? (ii) five years' time? and (iii) ten year's time?

We don't know what RPI will be in the future. All we can do is model the possible values of RPI.

3.4 Index-Linked Gilts (ILGs)

The first *index-linked gilt (ILG)* was issued by the UK Government in March 1981. Similar government bonds in other countries are Treasury Inflation-Protected Securities (TIPS), issued by the US Treasury since 1997, Obligations assimilables du Trésor Indexées sur l'inflation (OATi) issued by the L'Agence Trésor de France since 1998 and Real Return Bonds (RRB) issued by the Bank of Canada since 1991. Colloquially, index-linked bonds are called *linkers*.

As at 30 September 2013, the combined market value of ILGs is nearly £375 billion, compared to a total market value for conventional (fixed-interest) gilts of around £1 140 billion (Source: *DMO Quarterly Review for Jul-Sep 2013*). More information on the UK index-linked gilts, including their history, can be found at the UK Debt Management Office (DMO) website: `http://www.dmo.gov.uk/index.aspx?page=Gilts/Indexlinked`.

Example 3.19 The first ILG was the 2% Index-linked Treasury Stock 1996, with an *issue date* of 27 March 1981. For each £100 nominal, it paid:

- a coupon payment of £1 increased with RPI and paid on 16 March and 16 September of each year.
- a redemption payment of £100 increased with RPI and paid on 16 March 1996.

Calculating the money coupon and redemption payments for ILGs

For ILGs, the semi-annual coupon payments and redemption proceeds are adjusted in line with inflation as measured by the RPI. The rough idea is

$$\begin{aligned}\text{Money payment} = &\text{ Coupon or redemption payment} \\ &\times \frac{\text{Lagged RPI for the month that the payment is made}}{\text{Base RPI figure}}.\end{aligned}$$

For ILGs issued on or after 2005, the *base RPI figure* is the RPI that is 3 months before the issue date. In other words, the *time lag* is 3 months. The reason for introducing a time lag is that it is not possible to index payments to the RPI in the month in which they are paid. A coupon payment in January 2014 cannot be increased using the RPI in January 2014 since the latter figure is not known until after the coupon payment is made.

For ILGs issued before 2005, the time lag is 8 months. The reason was to allow a purchaser of a ILG to know exactly the amount of the next coupon payment. It was changed to a 3-month time lag to be more consistent with other governments' index-linked bonds and increase liquidity and transparency in the ILG market. Additionally, a 3-month time lag provides better inflation protection than an 8-month time lag.

Example 3.20 The $4\frac{1}{8}\%$ Index-linked Treasury Gilt 2030 had an issue date of 12 June 1992 and uses an 8-month time lag. Its base RPI figure is 135.1, the RPI for October 1991.

It pays coupons on 22 January and 22 July each year. The latest coupon payments are shown in Table 3.

Coupon payment date	22 Jan 2009	22 Jul 2009	22 Jan 2010	22 Jul 2010	22 Jan 2011
Lagged RPI date	May 2008	Nov 2008	May 2009	Nov 2009	May 2010
Lagged RPI	215.1	216.0	212.8	216.6	223.6
Coupon payment (£ per £100 nominal)	3.2838	3.2975	3.2487	3.3067	3.4135

Table 3: Recent coupon payments for the $4\frac{1}{8}\%$ Index-linked Treasury Gilt 2030.

The coupon payment made at 22 July 2011 is calculated as

$$\frac{4\frac{1}{8}}{2} \times \frac{\text{RPI(Jul 2011} - \{8 \text{ months}\})}{\text{Base RPI figure}} = 2.0625 \times \frac{\text{RPI(Nov 2010)}}{\text{RPI(Oct 1991)}} = 2.0625 \times \frac{226.8}{135.1} = 3.4624352.$$

The figure is rounded *down* to 4 decimal places, to give a coupon payment of £3.4624 per £100 nominal.

The calculation of the coupons and redemption payments is fixed upon the issuance of each ILG. ILGs with a 3-month time lag have a slightly more complicated calculation that that for ILGs with an 8-month time lag. The rounding convention is also different depending on when the ILG is issued. For example, for the 2% Index-linked Treasury Stock 2035 stock, the figure is rounded *to the nearest* 6 decimal places.

Exercise 3.21 Check the RPI figures and the coupon payments in Table 3.

Calculating the money yield on ILGs

Suppose I want to calculate the money yield on an ILG, assuming that I hold it until redemption. To do this I need the future money cashflows. However, I do not know what these will be, as they depend on future inflation.

What can I do to give me an idea of the possible values of the money yield?

Idea: Assume that future inflation is a chosen fixed rate, e.g. 2.5% per annum.

Method: Using the last published RPI figure, project RPI into the future at the chosen fixed rate. Using the projected RPI, I calculate the future coupon and redemption cashflows, which I can then use to calculate a money yield.

Example 3.22 Suppose today is 28 November 2013, and the last published RPI is RPI(Oct 2013) = 251.9. Assuming that RPI increases in the future at a constant rate of 2.5% per annum, the projected RPI for November 2013 is calculated as

$$\text{RPI(Nov 2013)} = \text{RPI(Oct 2013)} \times 1.025^{1/12} = 252.418873.$$

Additional projections are shown below.

RPI date	Oct 2013	Nov 2013	Dec 2013	Jan 2014	Feb 2014
Projected RPI for date	215.1	252.418873	252.938814	253.459827	253.981913

Example 3.23 Suppose today is 22 January 2013 and the last published RPI figure is RPI(Dec 2012)= 246.8.

Today's market price of £100 nominal of the $4\frac{1}{8}\%$ Index-linked Treasury Stock 2030 is £323.57, just after the payment of the January 2013 coupon. The $4\frac{1}{8}\%$ Index-linked Treasury Stock 2030 was issued in June 1992 and has an 8-month time lag. Its Base RPI figure is 135.1, the October 1991 RPI.

Coupon dates: 22 January, 22 July. Redemption date: 22 July 2030.

Calculate the **money** yield-to-redemption assuming that RPI will increase in the future at a constant effective rate of 2.5% per annum.

Solution To calculate the money yield-to-redemption we need to:

1. Calculate the future money coupon and redemption payments by projecting the last published RPI figure at the assumed inflation rate, and
2. Solve the equation-of-value for the annual money yield i.

Payment date	8-month lagged RPI date	8-month lagged RPI	Money payment
22 Jan 2013	N/A		-323.57
22 Jul 2013	Nov 2012	245.6	$\frac{4.125}{2} \times \frac{\text{RPI(Nov 2012)}}{\text{Base RPI}} = \frac{4.125}{2} \times \frac{245.6}{135.1}$
22 Jan 2014	May 2013	$246.8 \times 1.025^{\frac{5}{12}}$	$\frac{4.125}{2} \times \frac{\text{RPI(May 2013)}}{\text{Base RPI}} = \frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{5}{12}}}{135.1}$
22 Jul 2014	Nov 2013	$246.8 \times 1.025^{\frac{11}{12}}$	$\frac{4.125}{2} \times \frac{\text{RPI(Nov 2013)}}{\text{Base RPI}} = \frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{11}{12}}}{135.1}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
22 Jan 2030	May 2029	$246.8 \times 1.025^{16\frac{5}{12}}$	$\frac{4.125}{2} \times \frac{\text{RPI(May 2029)}}{\text{Base RPI}} = \frac{4.125}{2} \times \frac{246.8 \times 1.025^{16\frac{5}{12}}}{135.1}$
22 Jul 2030	Nov 2029	$246.8 \times 1.025^{16\frac{11}{12}}$	$\left(100 + \frac{4.125}{2}\right) \times \frac{\text{RPI(Nov 2029)}}{\text{Base RPI}}$ $= \left(100 + \frac{4.125}{2}\right) \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1}$

The equation of value is obtained from discounting each of the money payments back from the *payment date* to "today's" date of 22 January 2013 at the money yield i, and then summing them up.

$$\begin{aligned} 0 = -323.57 &+ \frac{4.125}{2} \times \frac{245.6}{135.1} \times v^{0.5} \\ &+ \frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{5}{12}}}{135.1} \times v^{1} \\ &+ \frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{11}{12}}}{135.1} \times v^{1.5} + \cdots + \frac{4.125}{2} \times \frac{246.8 \times 1.025^{16\frac{5}{12}}}{135.1} \times v^{17} \\ &+ \frac{4.125}{2} \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times v^{17.5} + 100 \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times v^{17.5}. \end{aligned}$$

To simplify the expression, look for a geometric series, i.e. a series of the form $1 + r + r^2 + \cdots$.

$$\begin{aligned} 323.57 = &\frac{4.125}{2} \times \frac{245.6}{135.1} \times v^{0.5} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \left(1.025^{\frac{5}{12}} v^1 + 1.025^{\frac{11}{12}} v^{1.5} \cdots 1.025^{16\frac{11}{12}} v^{17.5}\right) \\ &+ 100 \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times v^{17.5}. \end{aligned}$$

Taking out a factor of $1.025^{-\frac{1}{12}} v^{0.5}$ from the terms in the brackets, we obtain

$$\begin{aligned} 323.57 = &\frac{4.125}{2} \times \frac{245.6}{135.1} \times v^{0.5} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times 1.025^{-\frac{1}{12}} \times v^{0.5} \times \left(1.025^{0.5} v^{0.5} + 1.025^1 v^1 \cdots 1.025^{17} v^{17}\right) \\ &+ 100 \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times v^{17.5}. \end{aligned}$$

Define

$$1 + j := \left(\frac{1}{1.025v}\right)^{1/2} = \left(\frac{1+i}{1.025}\right)^{1/2},$$

in which i is the money yield. Then the equation of value can be written as

$$\begin{aligned} 323.57 = &\frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{(1+i)^{0.5}} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{1}{12}} \times (1+i)^{0.5}} \times \left(\frac{1}{1+j} + \frac{1}{(1+j)^2} \cdots \frac{1}{(1+j)^{34}}\right) \\ &+ 100 \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times \frac{1}{(1+i)^{17.5}}, \end{aligned}$$

i.e.

$$\begin{aligned} 323.57 = &\frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{(1+i)^{0.5}} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{1}{12}} \times (1+i)^{0.5}} \times a^{j}_{\overline{34|}} \\ &+ 100 \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times \frac{1}{(1+i)^{17.5}}. \end{aligned}$$

This is as far as we can reasonably simplify the equation of value. We can solve it using a computer or by hand using trial-and-error. We find $i = 2.187\%$ p.a.

Remark We can use different inflation assumptions.

- Assuming a constant future inflation rate of 2.0% p.a. results in a money yield of $i = 1.710\%$ p.a.

- Assuming a constant future inflation rate of 3.0% p.a. results in a money yield of $i = 2.665\%$ p.a.
- We could use different inflation rates for different periods, such as
  - 3.0% p.a. up to 22 January 2018, followed by
  - 2.5% p.a. from 23 January 2018 to the maturity date.

Calculating the real yield on ILGs

Now we turn to calculating the real yield. For this, I need to the real values of the future coupon and redemption proceeds.

To calculate the real yield-to-redemption,

1. Calculate the future money coupon and redemption payments by projecting the last published RPI figure at the assumed inflation rate, to calculate

$$\text{Money payment} = \text{Coupon or redemption payment} \times \frac{\text{RPI(Payment date - Time lag)}}{\text{Base RPI figure}}.$$

2. Adjust the money values to the real values at a chosen fixed date, e.g. adjusting them to today's purchasing power, we calculate

$$\text{Real payment} = \text{Money payment} \times \frac{\text{RPI(Today's date)}}{\text{RPI(Payment date)}}.$$

3. Solve the equation of value for the annual real yield i'.

Example 3.24 Suppose today is 22 January 2013 and the last published RPI figure is RPI(Dec 2012)= 246.8. The market price of £100 nominal of the $4\frac{1}{8}\%$ Index-linked Treasury Stock 2030 is £323.57, just after the payment of the January 2013 coupon.

Calculate the **real** yield-to-redemption assuming that RPI will increase in the future at a constant effective rate of 2.5% per annum.

Solution We have already calculated the money payments. The last published RPI figure is RPI(Dec 2012)= 246.8.

To calculate the real payments at "today's" date of 22 January 2013, we multiple each money payment by

$$\frac{\text{RPI(Jan 2013)}}{\text{RPI(Payment date)}} = \frac{\text{RPI(Dec 2012)} \times 1.025^{\frac{1}{12}}}{\text{RPI(Payment date)}} = \frac{246.8 \times 1.025^{\frac{1}{12}}}{\text{RPI(Payment date)}}.$$

Payment date	RPI(Payment date)	RPI(Jan 2013) / RPI(Payment date)	Real payment at Jan 2013 = Money payment $\times \frac{\text{RPI(Jan 2013)}}{\text{RPI(Payment date)}}$
22 Jan 2013	$246.8 \times 1.025^{\frac{1}{12}}$	1	-323.57×1
22 Jul 2013	$246.8 \times 1.025^{\frac{7}{12}}$	$\frac{1}{1.025^{0.5}}$	$\frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{1.025^{0.5}}$
22 Jan 2014	$246.8 \times 1.025^{1\frac{1}{12}}$	$\frac{1}{1.025^{1}}$	$\frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{5}{12}}}{135.1} \times \frac{1}{1.025^{1}}$
22 Jul 2014	$246.8 \times 1.025^{1\frac{7}{12}}$	$\frac{1}{1.025^{1.5}}$	$\frac{4.125}{2} \times \frac{246.8 \times 1.025^{\frac{11}{12}}}{135.1} \times \frac{1}{1.025^{1.5}}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
22 Jan 2030	$246.8 \times 1.025^{17\frac{1}{12}}$	$\frac{1}{1.025^{17}}$	$\frac{4.125}{2} \times \frac{246.8 \times 1.025^{16\frac{5}{12}}}{135.1} \times \frac{1}{1.025^{17}}$
22 Jul 2030	$246.8 \times 1.025^{17\frac{7}{12}}$	$\frac{1}{1.025^{17.5}}$	$\left(100 + \frac{4.125}{2}\right) \times \frac{246.8 \times 1.025^{16\frac{11}{12}}}{135.1} \times \frac{1}{1.025^{17.5}}$

The equation of value is obtained from discounting each of the real payments back from the *payment date* to "today's" date of 22 January 2013 at the real yield i', and then summing them up. Setting $v := 1/(1+i')$ and cancelling out some of the 1.025 indices gives

$$\begin{aligned} 0 = -323.57 &+ \frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{1.025^{0.5}} \times v^{0.5} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{1} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{1.5} \\ &\cdots + \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{17.5} \\ &+ 100 \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{17.5}. \end{aligned}$$

i.e.

$$\begin{aligned} 323.57 = &\frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{1.025^{0.5}} \times v^{0.5} \\ &+ \frac{4.125}{2} \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{0.5} \times \left(v^{0.5} + v^{1} + \cdots + v^{17}\right) \\ &+ 100 \times \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times v^{17.5}. \end{aligned}$$

The equation of value simplifies to

$$323.57 = \frac{4.125}{2} \times \frac{245.6}{135.1} \times \frac{1}{1.025^{0.5}} \times v^{0.5} + \frac{246.8}{135.1} \times \frac{1}{1.025^{\frac{7}{12}}} \times \left(\frac{4.125}{2} \times v^{0.5} \times a_{\overline{17}|}^{(2)i'} + 100 \times v^{17.5} \right).$$

We solve to find $i' = -0.305\%$ p.a.

Remark

The main selling point of an index-linked bond is that it protects the investor from price inflation. However, if we re-calculate the real yield using

- a constant inflation rate 2.0% p.a. then $i' = -0.285\%$ p.a.
- a constant inflation rate 3.0% p.a. then $i' = -0.325\%$ p.a.

The reason that the real yield is not invariant to the inflation assumption is due to the indexation time lag. The impact of the indexation time lag is seen in the last equation through the terms $\frac{1}{1.025^{0.5}}$ and $\frac{1}{1.025^{\frac{7}{12}}}$.

If there was no indexation time lag, then

- the real yield would be independent of future inflation. It would depend only on the market price, and
- the real yield would be known as soon as the investor buys the bond.

In practice, as there is an indexation time lag,

- the real yield, actual or anticipated, depends on future inflation as well as the market price, and
- the investor is exposed to some inflation risk. The longer the time lag, the greater the inflation risk to which the investor is exposed.

Exercise 3.25 Explain the meaning of a negative real yield.

Exercise 3.26 Would you invest in an ILG if it had a negative real yield? Explain your reasoning.

Break-even inflation rate (BEIR)

Index-linked bonds can also give an indication of the market consensus on future inflation. Suppose we have two bonds which are identical in characteristics, except that one is an index-linked bond and the other is a fixed interest bond, i.e.

- FI bond with annual coupon C p.a. payable m times a year and redeemed at $R\%$ in N years.
- IL bond with annual coupon C p.a. payable m times a year and redeemed at $R\%$ in N years.

Define

$$\{\text{Break-even inflation rate}\} := \{\text{Money yield on FI Bond}\} - \{\text{Real yield on IL Bond}\}.$$

The break-even inflation rate gives the market consensus on future inflation (ignoring issues such as the indexation time lag). It is the inflation rate at which the investor is indifferent between holding the FI bond and the IL bond.

Example 3.27 Suppose a FI bond with coupon $4\frac{1}{8}\%$ p.a. paid twice a year on 22 January and 22 July and is redeemed at par on 22 July 2030 gives a money yield of 4.2%. The real yield on the $4\frac{1}{8}\%$ Index-Linked Treasury Stock 2030 is 1.5% p.a.

The break-even inflation rate is $4.2 - 1.5 = 2.7\%$. If an investor believes:

- future rate of inflation will be lower at 2.5% p.a, then s/he should buy the FI bond. Why? Because the investor believes the money yield from the IL bond will be lower than from the FI bond.
- future rate of inflation will be higher at 3.0% p.a, then s/he should buy the IL bond. Why? Because the investor believes the money yield from the IL bond is higher.

4 Stochastic Interest Rate Models

The financial contracts that are of interest to actuaries are often very long term, and the rates of interest over the long term play a significant part in their calculations.

Up to now we have generally looked at models where interest rates have remained constant, or we have assumed that we know, in advance, what the interest rates will be (for example, we have assumed that the interest rate will be 10% per annum for the first 2 years and 8% per annum thereafter).

But, in reality, we cannot know the rate of interest in the future. And, in fact, the further into the future we go, the less certain we can be.

One solution is to use a Stochastic Interest Rate Model.

In essence, we model the uncertainty about future interest rates by assuming that future interest rates are random variables.

4.1 Model A

We wish to invest for 10 years starting at time 0, but are uncertain about future interest rates.

We will assume that interest rates will be constant during the years from time 0 to time 10, but that the rate might be 2% p.a, 4% p.a. or 6% p.a, each being equally likely.

Let i_t represent the interest earned over the time $[t-1, t)$, for $t = 1, 2, \ldots$, i.e. i_t is the annual effective interest rate in year t.

Suppose for $t = 1, 2, \ldots$,

$$\mathbb{P}_A[i_t = 2\%] = \mathbb{P}_A[i_t = 4\%] = \mathbb{P}_A[i_t = 6\%] = \frac{1}{3}.$$

Under Model A, the interest rate is the same every year for $t = 1, 2, \ldots$ so the expected value of the interest rate is

$$\mathbb{E}_A(\text{annual interest rate}) = \frac{1}{3}(0.02 + 0.04 + 0.06) = 0.04.$$

i.e. the expected interest rate is 4% p.a.

We can also calculate the variance using the formula

$$\begin{aligned}
\text{var}_A(i_t) &= \mathbb{E}_A\left((i_t - \mathbb{E}_A(i_t))^2\right) \\
&= \frac{1}{3}\left((0.02 - 0.04)^2 + (0.04 - 0.04)^2 + (0.06 - 0.04)^2\right) \\
&= 0.0002667. \\
\Rightarrow \text{sd}_A(i_t) &\approx 0.0163, \qquad \text{where } \text{sd}_A = \text{standard deviation under Model A.}
\end{aligned}$$

Accumulation under Model A

Now, suppose we had invested £1 for 10 years at time 0, how much will we have at the end of year 10?

We will have

$$S_{10} = \begin{cases} 1.02^{10} & \text{with probability } 1/3 \\ 1.04^{10} & \text{with probability } 1/3 \\ 1.06^{10} & \text{with probability } 1/3 \end{cases}$$

So the expected amount of money is:-

$$\mathbb{E}_A(S_{10}) = \frac{1}{3}[1.02^{10} + 1.04^{10} + 1.06^{10}] = 1.497$$

$$\mathrm{sd}_A(S_{10}) = 0.234$$

Note: This is not the same as $1.04^{10} = 1.480$ because the **expected accumulation** is not equal to the **accumulation at the expected rate of interest**.

Remark: Jensen's inequality

Let $f : \mathbb{R} \mapsto \mathbb{R}$ be a convex function ($f'' \geq 0$) then:

$$\mathbb{E}(f(X)) \geq f\left(\mathbb{E}(X)\right)$$

for all random variables X for which

$$\mathbb{E}(|X|) < \infty \text{ and } \mathbb{E}\left(|f(X)|\right) < \infty.$$

Here: $X = i$ and $f(x) = (1+x)^{10}$, $f''(x) = 10 \times 9 \times (1+x)^8 > 0$

$$\mathbb{E}(f(X)) = \mathbb{E}((1+i)^{10}) = 1.497 > 1.480 = (1 + \mathbb{E}(i))^{10}$$

Present Value under Model A

How much should we invest at time 0 to have £1 in 10 years' time ?

Answer: we should invest

$$A_{10} = \begin{cases} 1.02^{-10} & \text{with probability } 1/3 \\ 1.04^{-10} & \text{with probability } 1/3 \\ 1.06^{-10} & \text{with probability } 1/3 \end{cases}$$

$\mathbb{E}_A(A_{10}) = 0.685 \neq 1.04^{-10} = 0.676$
$\mathrm{sd}_A(A_{10}) = 0.107$

4.2 Model B

The interest rate each year will be

$$i_t = \begin{cases} 2\% \text{ p.a. with probability } 1/3 \\ 4\% \text{ p.a. with probability } 1/3 \\ 6\% \text{ p.a. with probability } 1/3 \end{cases}$$

The rate each year is **independent** of the rate in any other year. This is different from Model A because rates in different years may be different.

More formally, we say that the interest rate in year t is i_t, for $t = 1, 2, \ldots, 10$, and $\{i_t\}_{t=1}^{10}$ is a sequence of i.i.d random variables with the above distribution.

Note: i.i.d. means "independent and identically distributed"

Whilst it is still not realistic to assume that an interest rate could randomly jump from one year to the next to one of the three rates, it is perhaps more realistic than assuming that the rate will be constant throughout the 10-year term.

In **each** year: $\mathbb{E}_B(i_t) = 0.04$ and $\text{sd}_B(i_t) = 0.0163$, the same values as for Model A.

Accumulation under Model B

Consider £1 invested for 10 years at time 0.

How much is accumulated at the end of year 10 (i.e. after 10 years)?

It is

$$S_{10} = (1 + i_1)(1 + i_2) \cdots (1 + i_{10})$$

because at the end of the first year £1 is invested at rate i_1 and grows to $1 + i_1$. This is invested the following year at rate i_2, so the fund now grows to $(1 + i_1)(1 + i_2)$, etc.

$$\begin{aligned} \mathbb{E}_B\left(S_{10}\right) &= \mathbb{E}_B\left(\prod_{t=1}^{10}(1 + i_t)\right) \\ &= \prod_{t=1}^{10}\mathbb{E}_B(1 + i_t) \qquad \text{by independence} \\ &= \prod_{t=1}^{10} 1.04 = 1.04^{10} \approx 1.480, \end{aligned}$$

which is **not** the same value as for Model A.

Furthermore, we have

$$\mathbb{E}_B\left(S_{10}^2\right) = \prod_{t=1}^{10}\mathbb{E}_B\left((1 + i_t)^2\right) = \left(\mathbb{E}_B\left((1 + i_t)^2\right)\right)^{10},$$

since the interest rates in each year are independent. Since $\mathbb{E}\left(i_t^2\right) = \text{var}(i_t) + (\mathbb{E}(i_t))^2$, we obtain

$$\begin{aligned}\mathbb{E}_B\left((1+i_t)^2\right) &= \mathbb{E}_B\left(1+2i_t+i_t^2\right)\\ &= 1+2\mathbb{E}_B(i_t)+\text{var}_B(i_t)+\mathbb{E}_B(i_t)^2\\ &= 1+2\times 0.04+0.0163^2+0.04^2 = 1.081866\end{aligned}$$

and

$$\mathbb{E}_B\left(S_{10}^2\right) = 1.081866^{10} = 2.19651.$$

We therefore obtain for the variance of S_{10} under Model B,

$$\begin{aligned}\text{var}_B\left(S_{10}\right) &= \mathbb{E}_B\left(\left(S_{10}\right)^2\right) - \left(\mathbb{E}_B\left(S_{10}\right)\right)^2 = 2.19651 - 1.04^{20} = 0.005388\\ &\Rightarrow \text{sd}_B\left(S_{10}\right) = 0.0734 \quad (c.f \quad \text{sd}_A\left(S_{10}\right) = 0.234)\end{aligned}$$

Present Values under Model B

How much should we invest at time 0 in order to have 1 unit at the end of year 10?

$$A_{10} = (1+i_1)^{-1}(1+i_2)^{-1}\ldots(1+i_{10})^{-1}$$

$$(1+i_t)^{-1} = \begin{cases} 1.02^{-1} & \text{with probability } 1/3\\ 1.04^{-1} & \text{with probability } 1/3\\ 1.06^{-1} & \text{with probability } 1/3\end{cases}$$

Hence we can calculate $\mathbb{E}_B\left[(1+i_t)^{-1}\right] = 0.96178$ and $\mathbb{E}_B\left[(1+i_t)^{-2}\right] = 0.92524$. Then

$$\begin{aligned}\mathbb{E}_B\left[A_{10}\right] &= \prod_{t=1}^{10}\mathbb{E}_B\left[(1+i_t)^{-1}\right] \qquad \text{by independence}\\ &= 0.96178^{10}\\ &= 0.67726 \qquad (\text{c.f } \mathbb{E}_A(A_{10}) = 0.685)\end{aligned}$$

and

$$\mathbb{E}_B\left[(A_{10})^2\right] = \mathbb{E}_B\left[\prod_{t=1}^{10}(1+i_t)^{-2}\right] = \prod_{t=1}^{10}\mathbb{E}_B\left[(1+i_t)^{-2}\right] = 0.92524^{10} = 0.45977,$$

from which we can calculate

$$\text{var}_B\left[A_{10}\right] = 0.00109$$

and so

$$\text{sd}_B\left[A_{10}\right] = 0.033 \qquad (\text{c.f. } \text{sd}_A[A_{10}] = 0.107).$$

Example 4.1 Suppose $\{i_t\}_{t=1}^{\infty}$ are i.i.d. and each i_t has the following distribution:

$$i_t = \begin{cases} 0.02 & \text{with} & \text{probability} & 1/3\\ 0.05 & \text{with} & \text{probability} & 1/3\\ 0.08 & \text{with} & \text{probability} & 1/3\end{cases}$$

Denote by X the accumulated amount at time 2 of £2 invested at time 0 and £1 invested at time 1. Calculate $E[X]$ and $Var[X]$.

Solution

$$X = 2(1+i_1)(1+i_2) + (1+i_2) = (1+i_2)(3+2i_1)$$
$$E[i_t] = \frac{1}{3}(0.02 + 0.05 + 0.08) = 0.05$$
$$E[i_t^2] = \frac{1}{3}(0.02^2 + 0.05^2 + 0.08^2) = 0.0031$$
$$E[X] = E[1+i_2]\,E[3+2i_1] = 1.05(3 + 2(0.05)) = 3.255$$
$$E[X^2] = E[(1+i_2)^2]\,E[(3+2i_1)^2] = E[1+2i_2+i_2^2]\,E[9+12i_1+4i_1^2]$$
$$= (1 + 2(0.05) + 0.0031)(9 + 12(0.05) + 4(0.0031)) = 10.6034$$
$$\Rightarrow Var[X] = 10.6034 - 3.255^2 = 0.0084$$

4.3 Independent Annual Rates of Interest

Model

Let i_t be the annual rate of interest in year t, for $t = 1, 2, \ldots$. We assume that $\{i_t\}_{t=1}^{\infty}$ is a sequence of i.i.d. random variables. Let S_n be the accumulation at time n of £1 invested at time 0 and let A_n be the accumulation at time n of:

	£1 invested at time 0
+	£1 invested at time 1
+	$\vdots$
+	£1 invested at time $n-1$.

Therefore, A_n is the accumulation of a constant annuity of £1 payable in advance and we clearly have

$$S_n = (1+i_1)(1+i_2)\ldots(1+i_n) = \prod_{t=1}^{n}(1+i_t)\ .$$

And

$$A_n = \left\{ \begin{array}{r} (1+i_1)(1+i_2)\ldots(1+i_n) \\ + \quad (1+i_2)\ldots(1+i_n) \\ + \quad \vdots \\ + \quad (1+i_n) \end{array} \right\} = \sum_{t=1}^{n}\prod_{k=t}^{n}(1+i_k)$$

Given a distribution for i_t, and knowledge of the moments of i_t:

- can we find an explicit formula for the distributions of A_n and/or S_n? In general, no.
- can we calculate the moments of A_n and/or S_n? Yes

The Moments of S_n

We can calculate the first moment.

$$\begin{aligned}\mathbb{E}(S_n) = \mathbb{E}\left(\prod_{t=1}^{n}(1+i_t)\right) &= \prod_{t=1}^{n}\mathbb{E}\,(1+i_t) \quad \text{by independence} \\ &= \left(\mathbb{E}\,(1+i_t)\right)^n \quad \text{as identically distributed.}\end{aligned}$$

In general, the mth moment of S_n is given by

$$\begin{aligned}\mathbb{E}(S_n^m) = \mathbb{E}\left[\left(\prod_{t=1}^{n}(1+i_t)\right)^m\right] &= \mathbb{E}\left[\prod_{t=1}^{n}(1+i_t)^m\right] \\ &= \prod_{t=1}^{n}\mathbb{E}[(1+i_t)^m] = \left(\mathbb{E}[(1+i_t)^m]\right)^n.\end{aligned}$$

Hence all moments of S_n can be calculated if we know the distribution and moments of $\{i_t\}_t$.

The Moments of A_n

First Moment $E[A_n]$

$$\begin{aligned}E[A_n] &= E\left[\sum_{t=1}^{n}\prod_{k=t}^{n}(1+i_k)\right] \\ &= \sum_{t=1}^{n}\prod_{k=t}^{n}E[1+i_k] \qquad \text{(independence)} \\ &= \sum_{t=1}^{n}(E[1+i_1])^{n-t+1} \qquad \text{(identically distributed)} \\ &= \sum_{t=1}^{n}(E[1+i_1])^t = \sum_{t=1}^{n}(1+E[i_1])^t \\ &= \ddot{s}_{\overline{n}|} \quad at \quad i = E[i_1].\end{aligned}$$

Second Moment $E[A_n^2]$

$$A_n = (1+i_n)(1+A_{n-1})$$

As $i_1, i_2, \ldots, i_{n-1}, i_n$ are independent random variables, and A_{n-1} is a function of $i_1, i_2, \ldots, i_{n-1}$ only, then A_{n-1} and i_n are independent, and hence so are $(1+A_{n-1})$ and $(1+i_n)$. Thus

$$\begin{aligned}E[A_n^2] &= E[(1+i_n)^2(1+A_{n-1})^2] = E[(1+i_n)^2]E[(1+A_{n-1})^2] \\ &= E[1+2i_n+i_n^2]\,E[1+2A_{n-1}+A_{n-1}^2] \\ &= (1+2E[i_n]+E[i_n^2])\,(1+2E[A_{n-1}]+E[A_{n-1}^2]).\end{aligned}$$

The only unknown part of this is $E[A_{n-1}^2]$ but we know $E[A_1^2]$ because $A_1 = (1+i)$ so we can calculate the others recursively, using $E[A_1^2]$ we calculate $E[A_2^2]$, etc ....

The mth Moment

We use a similar recursive method.

$$A_n^m = (1+i_n)^m(1+A_{n-1})^m$$

$$\begin{aligned}
E[A_n^m] &= E[(1+i_n)^m]\, E[(1+A_{n-1})^m], \quad \text{by independence} \\
&= E[(1+i_n)^m]\, E\left[\sum_{k=0}^{m} \frac{m!}{k!(m-k)!}(A_{n-1})^k\right], \quad \text{(binomial expansion)} \\
&= E[(1+i_n)^m] \sum_{k=0}^{m} \frac{m!}{k!(m-k)!} E[(A_{n-1})^k]
\end{aligned}$$

We can calculate $E[A_n^m]$ recursively for $n = 2, 3, \ldots$ having first calculated $E[A_{n-1}^k]$ for $k = 1, 2, \ldots, m$.

Note that we have looked at accumulations, A_n, S_n in this section, but we could just as easily have looked at present values.

Example 4.2 The amount we need to invest now to pay £1 at times $0, 1, 2, \ldots, n-1$ is

$$1 + (1+i_1)^{-1} + (1+i_1)^{-1}(1+i_2)^{-1} + \ldots + (1+i_1)^{-1}\ldots(1+i_{n-1})^{-1}$$

$$= \sum_{t=0}^{n-1} \prod_{k=0}^{t} (1+i_k)^{-1} \qquad \text{(with } i_0 = 0\text{)}$$

4.4 The Independent Lognormal Model

In this section we assume that $(1+i_t)$ is lognormally distributed with parameters (μ, σ^2),

$$\begin{aligned}
&(1+i_t) \sim \text{logN}(\mu, \sigma^2), \quad \text{for } t = 1, 2, \ldots \\
\Leftrightarrow &\ln(1+i_t) \sim N(\mu, \sigma^2).
\end{aligned}$$

We can then use standard results (based on MGF of $N(\mu, \sigma^2)$):

$$E[1+i_t] = \exp\left(\mu + \frac{1}{2}\sigma^2\right)$$

$$E[(1+i_t)^m] = \exp\left(\mu m + \frac{1}{2}m^2\sigma^2\right)$$

$$Var[1+i_t] = \exp\left(2\mu + \sigma^2\right)\left[\exp\left(\sigma^2\right) - 1\right]$$

We are still assuming that $\{i_t\}_{t\geq 1}$ are i.i.d. This is a special case of the model in the previous section.

Note that $(1+i_t)$ is distributed over $(0,+\infty)$ so, although we can have a negative interest rate i_t, we cannot have negative accumulations.

Reminder
(1) $X \sim N(\mu,\sigma^2)$ then $\frac{X-\mu}{\sigma} \sim N(0,1)$.
(2) If $X_1,\ldots,X_n$ is i.i.d. and $X_1 \sim N(\mu,\sigma^2)$ then $\sum_{i=1}^n X_i \sim N(n\mu, n\sigma^2)$.
(3) Given $X \sim N(\mu,\sigma^2)$ and real-valued constants $a \neq 0$ and b, then $aX+b \sim N(a\mu+b, a^2\sigma^2)$.

Mathematical Properties

$$S_n = \prod_{t=1}^{n}(1+i_t) \implies \ln(S_n) = \sum_{t=1}^{n}\ln(1+i_t) \sim N(n\mu, n\sigma^2)$$

Hence $S_n \sim \text{logN}(n\mu, n\sigma^2)$.

Furthermore

$$\begin{aligned} E[S_n] &= \exp\left(n\mu + \frac{1}{2}n\sigma^2\right) \qquad \text{(MGF again)} \\ Var[S_n] &= \exp\left(2n\mu + n\sigma^2\right)\left[\exp\left(n\sigma^2\right) - 1\right] \end{aligned}$$

Let X_n be the present value of £1 payable at time n, then

$$X_n = \prod_{t=1}^{n}(1+i_t)^{-1}.$$

As

$$\ln[(1+i_t)^{-1}] = -\ln(1+i_t) \sim N(-\mu,\sigma^2),$$

we find $X_n \sim \text{logN}(-n\mu, n\sigma^2)$.

Conclusion: for the independent lognormal model, the accumulation and the present value of a single payment have a known distribution.

Example 4.3 Assume the $\{i_t\}_{t=1}^{\infty}$ are i.i.d. and that $(1+i_t) \sim \text{logN}(\mu,\sigma^2)$. Given that $E[i_t] = 0.05$ and $Var[i_t] = 0.088^2$,

1. Calculate μ and σ^2.
2. Calculate the probability $S_{12} > 1.8$.
3. Calculate $E(A_2)$ and $Var(A_2)$.

Solution
(i)

$$E[1+i_t] = \exp\left(\mu + \frac{1}{2}\sigma^2\right) = 1.05$$

$$Var[i_t] = Var[1+i_t] = \exp\left(2\mu+\sigma^2\right)\left[\exp\left(\sigma^2\right)-1\right] = 0.088^2$$

$$\Rightarrow 0.088^2 = 1.05^2\left[e^{\sigma^2}-1\right] \quad \Rightarrow \quad \sigma^2 \approx 0.007000$$

$$\mu + \frac{1}{2}\sigma^2 = \ln(1.05) \quad \Rightarrow \quad \mu \approx 0.04529$$

(ii) $S_{12} \sim \text{logN}(12\mu, 12\sigma^2)$.

$$\mathbb{P}[S_{12} > 1.8] = \mathbb{P}[\ln(S_{12}) > \ln 1.8] = \mathbb{P}\left[\underbrace{\frac{\ln(S_{12})-12\mu}{\sqrt{12\sigma^2}}}_{\sim N(0,1)} > \underbrace{\frac{\ln 1.8 - 12\mu}{\sqrt{12\sigma^2}}}_{\approx 0.153}\right]$$

$$= 1 - \Phi(0.153) \approx 1 - \Phi(0.15) \approx 1 - 0.5596 = 0.4404$$

where Φ is the distribution function for the standard normally distributed random variable.

(iii)

$$E[A_2] = \ddot{s}_{\overline{2}|}^{0.05} = 2.1525$$

$$A_2 = (1+i_2)(1+A_1) = (1+i_2)(2+i_1) \Rightarrow A_2^2 = (1+i_2)^2(2+i_1)^2$$

$$\begin{aligned}
\Longrightarrow E[A_2^2] &= E[(1+i_2)^2]E[(2+i_1)^2] \\
&= \left[Var(1+i_2) + (E[1+i_2])^2\right] E\left[4+4i_1+i_1^2\right] \\
&= \left[0.088^2 + 1.05^2\right]\left[4+4(0.05)+Var[i_1]+\left(E[i_1]^2\right)\right] \\
&\approx 4.6744
\end{aligned}$$

$\Rightarrow Var[A_2] = E[A_2^2] - (E[A_2])^2 \approx 0.04114.$

5 Nominal Interest Rates

Throughout this section the time unit will be one year.

5.1 Nominal Rates of Interest

Say you put £100 into a savings account at a bank for a year. Which effective interest rate would you rather have

- 6% per annum?
- 2.9563% per half-year?

Consider a rate of interest i per annum.

- **Effective** interest means that interest of £i is paid at the year end on £1 invested at the start of the year.
- **Nominal** interest of $i^{(m)}$ per annum compounding m times a year means that £$\frac{i^{(m)}}{m}$ is paid at the end of each fraction $\frac{1}{m}$ of a year on £1 invested at the start of the year.

Example 5.1 The following are equivalent:

Effective Rate (p.a.)	**Nominal Rate (p.a.)**
$i = 0.06$	$i^{(2)} = 0.059126$
$i = 0.12$	$i^{(12)} = 0.113866$
$i = 0.082432$	$i^{(4)} = 0.08$
$i = 0.092025$	$i^{(2)} = 0.09$

$i^{(m)}$ per annum convertible m^{thly} or m times per year

is identical to

an effective rate of interest $\frac{i^{(m)}}{m}$ per $\frac{1}{m}$ year.

It is usually easier to change to the appropriate time unit and use the effective rate of interest, but you might still come across this (pre-calculator) notation. It can be useful for remembering some of the annuity formula.

The relationship between the effective interest rate i p.a. and the nominal interest rate $i^{(m)}$ p.a. is

$$1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m.$$

Exercise 5.2 The effective rate of interest is 3.5% per annum. Express this as the equivalent annual nominal rate of interest:

1. convertible quarterly.
2. convertible 12 times a year.

Exercise 5.3 The nominal rate of interest per annum convertible half-yearly is 5% per annum. Express this as the equivalent annual

1. effective rate of interest.
2. nominal rate of interest convertible monthly.

5.2 Nominal Rates of Discount

Recall that $d = 1 - v$ is the effective rate of discount per annum. It could be interpreted as the amount of interest payable at the start of the year that is equivalent to i payable at the end of the year.

Example 5.4 Generally, bank or savings accounts pay interest at the end of a set time period. However, Santander, a retail bank, launched a 3-year savings account in 2011 that paid interest upfront. No withdrawals were permitted during the 3 year time period.

The *nominal discount rate* $d^{(m)}$ represents interest of $\frac{d^{(m)}}{m}$ paid at the start of each $\frac{1}{m}$ of a year.

$d^{(m)}$ per annum convertible m^{thly} or m times per year

is identical to

an effective rate of discount $\frac{d^{(m)}}{m}$ per $\frac{1}{m}$ year.

The relationship between the effective discount rate d p.a. and the nominal discount rate $d^{(m)}$ p.a. is

$$\boxed{1 - d = \left(1 - \frac{d^{(m)}}{m}\right)^m.}$$

Example 5.5 The nominal discount rate convertible quarterly is 8% per annum. Express this as the equivalent annual:

1. effective discount rate. 2. effective interest rate.

Solution We are told $d^{(4)} = 0.08$ p.a. Hence

$$1 - d = \left(1 - \frac{0.08}{4}\right)^4 \qquad \Rightarrow \qquad d = 1 - 0.98^4 = 0.077632.$$

Then as $v = 1 - d = 0.98^4$, we find

$$i = \frac{1}{0.98^4} - 1 = 0.084166.$$

Exercise 5.6 The effective rate of discount is 4% per annum. Express this as the equivalent annual nominal rate of discount:

1. convertible quarterly.
2. convertible half-yearly.

Exercise 5.7 The nominal rate of discount per annum convertible monthly is 6% per annum. Express this as the equivalent annual

1. effective rate of discount.
2. nominal rate of discount convertible quarterly.

Table 4: Equivalent Payments

	0	$1/p$	$2/p$	$3/p$	...	$(p-1)/p$	1
(1)	d						
(2)	$d^{(p)}/p$	$d^{(p)}/p$	$d^{(p)}/p$	$d^{(p)}/p$	...	$d^{(p)}/p$	
(3)		$i^{(p)}/p$	$i^{(p)}/p$	$i^{(p)}/p$	...	$i^{(p)}/p$	$i^{(p)}/p$
(4)							i

Example 5.8 £2500 is invested in a savings account. The nominal rate of interest per annum convertible monthly for the first 3 months is 24% and the nominal rate of interest convertible quarterly for the next 9 months is 16%. How much is in the account at the end of the year?

Solution:

$$\begin{aligned}
&\implies 2500\left(1+\frac{i^{(12)}}{12}\right)^3\left(1+\frac{i^{(4)}}{4}\right)^3\\
&\implies 2500\left(1+\frac{0.24}{12}\right)^3\left(1+\frac{0.16}{4}\right)^3\\
&\implies 2500(1+0.02)^3(1+0.04)^3\\
&\implies 2984.29
\end{aligned}$$

Annuities payable mthly

- $a_{\overline{n}|}^{(m)}$: The value at time zero of a level annuity payable $m$thly in arrears at the rate of 1 per unit time over the time interval $[0, n]$.
- $a_{\overline{n}|}^{(m)}$: The value at time zero of a level annuity which pays $1/m$ in arrears m times per year for a period of n years.
- $s_{\overline{n}|}^{(m)}$: The value at time n of a level annuity payable $m$thly in arrears at the rate of 1 per unit time over the time interval $[0, n]$.

Useful results

$$
\begin{aligned}
a^{(m)}_{\overline{n}|} &= \sum_{t=1}^{nm} \frac{1}{m} v^{\frac{t}{m}} = \frac{1-v^n}{i^{(m)}} \\
\ddot{a}^{(m)}_{\overline{n}|} &= \sum_{t=1}^{nm} \frac{1}{m} v^{\frac{t-1}{m}} = \frac{1-v^n}{d^{(m)}} \\
a^{(m)}_{\overline{\infty}|} &= \frac{1}{i^{(m)}} \\
\ddot{a}^{(m)}_{\overline{\infty}|} &= \frac{1}{d^{(m)}} \\
s^{(m)}_{\overline{n}|} &= \sum_{t=1}^{nm} \frac{1}{m} (1+i)^{n-\frac{t}{m}} = \frac{(1+i)^n - 1}{i^{(m)}} \\
\ddot{s}^{(m)}_{\overline{n}|} &= \sum_{t=1}^{nm} \frac{1}{m} (1+i)^{n-\frac{t-1}{m}} = \frac{(1+i)^n - 1}{d^{(m)}}
\end{aligned}
$$

Exercise 5.9 Calculate the present value as at 1 January 2014 of the payments of £100 made on the first day of each quarter during the calendar years 2014 and 2015, if the effective rate of interest is 8% per annum.

Exercise 5.10 An investor who pays income tax at 20% on all their income buys, at issue, a 25-year bond to be redeemed at par bearing semi-annual (twice per year) coupons of 8% per annum. What price will the investor pay per £100 nominal to obtain a redemption yield after income tax of 9% per annum effective?

Annuities payable mthly where $\mathbf{m < 1}$

- $m < 1$: An annuity that pays less frequently than annually. For example, $m = 0.5$ means payments are made once in 2 years i.e. biennially.

Exercise 5.11 Assuming that the rate of interest is i per annum, what is the value at time 0 of a series of 7 payments, each of amount 8, made at time 4, 8, . . . , 28?

Example 5.12 Suppose that k and r are integers greater than 1. Consider a series of payments, each of amount X, due at times r, $2r$, $3r$, $\ldots$, kr. What is the value of this series at time 0 on the basis of an effective interest rate i per unit time?

Solution: The situation can be illustrated by replacing the payment of X due at time r by a series of r payments, each of amount Y, due at times $1, 2, \ldots, r$, where Y is chosen such that the future value at time r is equivalent to the single payment they replace i.e. X at time r. This means that $Y s_{\overline{r}|} = X$.

Hence the value of the annuity is:

$$\left(\frac{X}{s_{\overline{r}|}}\right) a_{\overline{kr}|}.$$

Exercise 5.13 A company expects to receive £10,000 from a customer. The amount will be split into ten equal instalments. The first instalment is due in one years' time and the remaining payments will be made at three-year intervals thereafter. Calculate, to the nearest £, the present value of these future receipts at an effective rate of 7% per annum.

6 Force of Interest and Continuously-paid Cash Flows

6.1 The Force of Interest

A sum of money $A(0)$ is invested at time 0 in a bank account and it grows continuously. At time t the amount of money in the bank account is $A(t)$.

Definition 6.1

The force of interest at time t is

$$\delta(t) = \frac{1}{A(t)} \frac{dA(t)}{dt} \tag{6.1}$$

Remark: From equation (6.1), we see

$$\delta(t) = \frac{d}{dt}(\log(A(t))$$
$$\Rightarrow \int_{t_1}^{t_2} \delta(t)dt = \log A(t_2) - \log A(t_1)$$

Rearranging, we find the important general formula:

$$A(t_2) = A(t_1) \exp\left(\int_{t_1}^{t_2} \delta(s)ds\right). \tag{6.2}$$

For example, set $t_1 = 0$ and $t_2 = t$. Then

$$A(t) = A(0) \exp\left(\int_0^t \delta(s)ds\right).$$

If we know $\delta(s)$ from $s = 0$ to $s = t$, then we can calculate

- the accumulation of $A(0)$ to time t.
- the present value of $A(t)$ at time 0.

For example, suppose the force of interest is constant, $\delta(t) = \delta$ for all $t \geq 0$ the accumulation of $A(0) = 100$ from time 0 to:

- time 1 is $A(1) = 100 \exp\left(\int_0^1 \delta ds\right) = 100e^{\delta}$.
- time 2 is $A(2) = 100 \exp\left(\int_0^2 \delta ds\right) = 100e^{2\delta}$.
- time t is $A(t) = 100 \exp\left(\int_0^t \delta ds\right) = 100e^{\delta t}$.

Suppose we had been given an annual effective of rate of interest i. Then at time t, the accumulation of $A(0) = 100$ is

$$A(t) = 100(1 + i)^t.$$

Therefore

$$A(t) = 100e^{\delta t} = 100(1+i)^t$$

i.e

$$e^{\delta} = 1 + i.$$

Exercise 6.1 Given a constant force of interest $\delta(t) = 5\%$ p.a, for all $t \geq 0$, and assuming time is measured in years, calculate

1. the PV at time $t = 0$ of £150 paid at time $t = 1$.
2. the PV at time $t = 0.5$ of £200 paid at time $t = 3.25$.
3. the accumulation at time $t = 4$ of £50 paid at time $t = 1$.

Exercise 6.2 Write down a formula for the annual effective interest rate i in terms of the constant force of interest δ.

More generally,

- a constant force of interest δ p.a.
- an effective rate of interest i p.a.
- a nominal rate of interest $i^{(p)}$ p.a. convertible p times per year
- a nominal rate of discount $d^{(q)}$ p.a. convertible q times per year
- an effective rate of discount d p.a.

are all equivalent if and only if the following holds:

$$e^{\delta} = 1 + i = \left(1 + \frac{i^{(p)}}{p}\right)^{p} = \left(1 - \frac{d^{(q)}}{q}\right)^{-q} = (1-d)^{-1}.$$

Exercise 6.3 Write down a formula for the annual discount rate v in terms of the constant force of interest δ.

Exercise 6.4 Given that the annual effective rate of interest is 12%, calculate d, $d^{(4)}$, $d^{(12)}$, $i^{(4)}$, $i^{(12)}$ and δ.

Another approach to the constant force of interest

The force of interest δ can also be interpreted as a nominal rate of interest compounded infinitely often, with

$$\delta = \lim_{m \to \infty} i^{(m)}.$$

The reasoning is as follows. Over one year, £1 accumulates to

- $1 + i$, given a nominal rate of interest $i^{(1)} = i$ convertible yearly.

- $(1+\frac{i^{(4)}}{4})^4$, given a nominal rate of interest $i^{(4)}$ convertible quarterly.
- $(1+\frac{i^{(365)}}{365})^{365}$, given a nominal rate of interest $i^{(365)}$ convertible daily.
- $(1+\frac{i^{(8760)}}{8760})^{8760}$, given a nominal rate of interest $i^{(8760)}$ convertible hourly.
- $(1+\frac{i^{(m)}}{m})^m$, given a nominal rate of interest $i^{(m)}$ convertible $\frac{1}{m}$-ly.

As $m \to \infty$ (i.e. continuous compounding), $i^{(m)} \to \delta$ and

$$\lim_{m\to\infty}\left(1+\frac{\delta}{m}\right)^m = e^{\delta}$$

Example 6.5 Find the accumulated amount at time 20 of a cash flow of £1 000 at time 0, assuming a force of interest $\delta(t)$ p.a. given by

$$\delta(t) = 0.06 + 0.0005t - 0.00005t^2.$$

Solution: Accumulated value at time 20 is

$$\begin{aligned}
&1\,000\exp\left(\int_0^{20}\delta(s)ds\right)\\
=&1\,000\exp\left(\int_0^{20}(0.06+0.0005s-0.00005s^2)ds\right)\\
=&1\,000\exp\left(\left[0.06s+\frac{0.0005}{2}s^2-\frac{0.00005}{3}s^3\right]_{s=0}^{s=20}\right)\\
=&1\,000\times\exp\left(\frac{3.5}{3}\right)\\
=&3\,211.27.
\end{aligned}$$

Example 6.6 Find the accumulated amount at time 20 of a cash flow of £1 000 at time 0 and a further £1 000 at time 10, assuming a force of interest $\delta(t)$ p.a. given by

$$\delta(t) = 0.06 + 0.0005t - 0.00005t^2.$$

Solution:

Time	0	10
Cashflow	1 000	1 000
AV of cashflow at time 20	$1,000\exp\left(\int_0^{20}\delta(s)ds\right)$ $=1\,000\times\exp\left(\frac{3.5}{3}\right)$	$1\,000\exp\left(\int_{10}^{20}\delta(s)ds\right)$ $=1\,000\times\exp\left(\frac{1.675}{3}\right)$

Hence the accumulated value at time 20 is

$$1\,000\times\exp\left(\frac{3.5}{3}\right)+1\,000\times\exp\left(\frac{1.675}{3}\right)=£4\,959.03$$

Exercise 6.7 Calculate the present value at time 0 of a cashflow of 100 at time 4, and 1000 at time 10 assuming a force of interest per time unit given as follows:

$$\delta = \begin{cases} 0.04 & 0 \leq t < 5 \\ 0.14 - 0.02t & 5 \leq t \end{cases}$$

6.2 Continuously-paid cashflows

Let $M(t)$ denote the total payment made between time 0 and time t. Then, the rate of payment is

$$\rho(s) = M^{'}(s), \text{ for all } s \geq 0.$$

For $0 \leq t_1 < t_2$, the total payment between time t_1 and time t_2 is

$$M(t_2) - M(t_1) = \int_{t_1}^{t_2} M^{'}(s)ds = \int_{t_1}^{t_2} \rho(s)ds.$$

Example 6.8 Start with zero money in the bank, i.e. $M(0) = 0$, and assume no interest. Put money in the bank from time 0 at the constant rate of £500 p.a, i.e $\rho(t) = £500$ p.a. for $t \geq 0$.

After one year, there is £500 in the bank, i.e. $M(1) = 500$. After 1.5 years, there is £750 in the bank, i.e. $M(1.5) = 750$. In general,

$$M(t) = M(0) + \int_0^t \rho(s)ds = 0 + \int_0^t 500ds = 500t.$$

Present Values and Accumulated Values

For times s and t with $s < t$, denote by

- $V(s,t)$ the present value at time s of 1 unit paid at time t.
- $A(s,t)$ the accumulation at time t of 1 unit paid at time s.

Remark: $V(s,t) = \frac{1}{A(s,t)}$.

Example 6.9 For a constant force of interest δ per annum,

- $V(s,t) = e^{-\delta(t-s)}$.
- $A(s,t) = e^{\delta(t-s)}$.

Present Values

Consider a cashflow paid continuously at the rate

$$\rho(t) \quad \text{p.a.}$$

for T years starting at time 0. Partition the time interval $[0, T]$ into tiny disjoint segments, each of length dt. Consider one of these tiny segments $[t, t + dt)$.

If dt is very small, then the total payment between time t and time $t + dt$ is

$$M(t + dt) - M(t) \approx \rho(t)dt.$$

We treat the amount paid between t and $t+dt$ as paid exactly at time t, i.e. we treat the continuous payment made over $[t, t + dt)$ as a discrete cashflow $\rho(t)dt$ paid at time t.

The PV at time 0 of the discrete cashflow is $V(0, t)\rho(t)dt$.

Repeat for all the tiny segments intervals, then sum them to find the total PV at time 0. Letting $dt \to 0$, we find in the limit that the present value at time 0 of the entire cash flow between time 0 and time T is

$$\int_0^T \rho(t)V(0, t)dt = \int_0^T \rho(t) \exp\left(-\int_0^t \delta(u)du\right) dt.$$

Example 6.10 Find the present value at time 0 of a cashflow paid continuously at the constant rate £500 p.a. from time 0 to time 10, assuming a constant force of interest δ p.a.

Solution: As $\rho(t) = 500$ and $V(0, t) = e^{-\delta t}$,

$$\begin{aligned}
\text{PV } (0) &= \int_0^{10} \rho(t)V(0, t)dt \\
&= \int_0^{10} 500e^{-\delta t}dt \\
&= 500\left[\frac{e^{-\delta t}}{-\delta}\right]_{t=0}^{t=10} \\
&= 500\frac{1 - e^{-10\delta}}{\delta} \\
&= 500\frac{1 - v^{10}}{\delta}
\end{aligned}$$

with $v = e^{-\delta}$.

Exercise 6.11 Assume time is measured in years.

1. Given a constant force of interest $\delta = 3\%$ p.a. calculate the PV at time $t = 0$ of a cashflow paid continuously from time $t = 0$ to time $t = 2$ at the rate £10 000 p.a.
2. Given a constant effective rate of interest $i = 6\%$ p.a, calculate the PV at time $t = 0$ of a cashflow paid continuously from time $t = 0$ to time $t = 2$ at the rate £10 000 p.a.
3. Given a constant force of interest $\delta = 3\%$ p.a, calculate the PV at time $t = 0$ of a cashflow paid continuously from time $t = 0$ to time $t = 2$ at the rate $\rho(t) = e^{-0.05t}$ p.a, for all $t \in [0, 2]$.

More generally, consider a cash flow paid continuously at the rate $\rho(t)$ p.a. for T years starting at time S. Partition the time interval $[S, T]$ into tiny disjoint segments, each of length dt. Consider one of these tiny segments $[t, t + dt)$.

If dt is very small, then the total payment between time t and time $t+dt$ is

$$M(t+dt) - M(t) \approx \rho(t)dt.$$

We treat the amount paid between t and $t+dt$ as paid exactly at time t, i.e. we treat the continuous payment made over $[t, t+dt)$ as a discrete cashflow $\rho(t)dt$ paid at time t.

The PV at time r of the discrete cashflow is $V(r,t)\rho(t)dt$.

Repeat for all the tiny segments intervals, then sum them to find the total PV at time r. Letting $dt \to 0$, we find in the limit that the present value at time r of the entire cash flow between time S and time T is

$$\int_S^T \rho(t)V(r,t)dt = \int_S^T \rho(t) \exp\left(-\int_r^t \delta(u)du\right) dt.$$

Example 6.12 Find the present value at time 2 of a cashflow paid continuously at the constant rate £1 000 p.a. from time 5 to time 20, assuming a constant force of interest δ p.a.

Solution: As $\rho(t) = 1\,000$ and $V(2,t) = e^{-\delta(t-2)}$,

$$\begin{aligned}
\text{PV } (2) &= \int_5^{20} \rho(t)V(2,t)dt \\
&= \int_5^{20} 1\,000e^{-\delta(t-2)}dt \\
&= 1\,000\left[\frac{e^{-\delta(t-2)}}{-\delta}\right]_{t=5}^{t=20} \\
&= 1\,000\frac{e^{-3\delta} - e^{-18\delta}}{\delta} \\
&= 1\,000v^3\frac{1-v^{15}}{\delta}
\end{aligned}$$

with $v = e^{-\delta}$.

Exercise 6.13 Assume a constant force of interest $\delta = 3\%$ p.a. Calculate the present value at time $t = 1$ of a cashflow paid continuously at the constant rate £10 000 p.a. from time $t = 3$ to time $t = 14$.

Accumulated Values

For accumulated values, we use exactly the same approach.

Consider a cash flow paid continuously at the rate $\rho(t)$ p.a. for T years starting at time S. Partition the time interval $[S,T]$ into tiny disjoint segments, each of length dt. Consider one of these tiny segments $[t, t+dt)$. The total payment between time t and time $t+dt$ is approximately $\rho(t)dt$.

The AV at time T of the discrete cashflow is $A(t,T)\rho(t)dt$.

Repeat for all the tiny segments intervals, then sum them to find the total PV at time T. Letting $dt \to 0$, we find in the limit that the accumulated value at time T of the entire cash flow between time S and time T is

$$\int_S^T \rho(t)A(t,T)dt = \int_S^T \rho(t) \exp\left(\int_t^T \delta(u)du\right) dt$$

Example 6.14 Find the accumulated value at time 10 of a cashflow paid continuously at the rate £1 000 p.a. from time 0 to time 10, assuming a constant force of interest δ p.a.

Solution: As $\rho(t) = 1\,000$ and $A(t,10) = e^{\delta(10-t)}$,

$$\begin{aligned}
\text{AV } (10) &= \int_0^{10} \rho(t)A(t,10)dt \\
&= \int_0^{10} 1\,000e^{\delta(10-t)}dt \\
&= 1\,000\left[\frac{e^{\delta(10-t)}}{-\delta}\right]_{t=0}^{t=10} \\
&= 1\,000\frac{e^{10\delta}-1}{\delta} \\
&= 1\,000\frac{(1+i)^{10}-1}{\delta}
\end{aligned}$$

with $1 + i = e^{\delta}$.

Exercise 6.15 Assume a constant force of interest $\delta = 3\%$ p.a. Calculate the accumulated value at time $t = 14$ of a cashflow paid continuously at the constant rate £10 000 p.a. from time $t = 3$ to time $t = 14$.

Exercise 6.16 Assume a constant force of interest $\delta = 4\%$ p.a. Calculate the accumulated value at time $t = 12$ of a cashflow paid continuously at the constant rate £100 p.a. from time $t = 3$ to time $t = 5$.

Exercise 6.17 The force of interest is

$$\delta(t) = \begin{cases} 0.04 & \text{for } 0 \le t < 5 \\ 0.03 & \text{for } t \ge 5. \end{cases}$$

Calculate the present value at time $t = 2$ of a payment stream paid continuously from $t = 5$ to $t = 10$ at a rate of £50 per annum.

Different types of cashflows

The accumulation at time $T > 0$ of

1. one discrete payment $C(t_1)$ made at time t_1 is

$$A(T) = C(t_1) \exp\left(\int_{t_1}^T \delta(s)ds\right).$$

2. n discrete payments $C(t_k)$ made at times t_k, for $k = 1, \ldots, n$ is

$$A(T) = \sum_{k=1}^{n} C(t_k) \exp\left(\int_{t_k}^{T} \delta(s)ds\right).$$

3. a continuous payment paid at rate $\rho(t)$ for all $t \in [S, T]$, for some $S < T$, is

$$A(T) = \int_{S}^{T} \rho(t) \exp\left(\int_{t}^{T} \delta(u)du\right) dt.$$

International actuarial notation for continuously-paid cashflows

Assume a constant force of interest δ p.a.

Constant annuity: constant rate of payment

Denote by $\bar{a}_{\overline{n}|}$ the PV at the start of a cashflow which is paid continuously for n years and has a constant rate of payment 1 unit p.a.

Important formula:

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta}.$$

Example 6.18 The PV at time 0 of a cashflow paid continuously at the rate £1 000 p.a. from time 0 to time 10 years, assuming a force of interest of 5% p.a, is

$$1\,000\bar{a}_{\overline{10}|} = 1\,000\frac{1 - e^{-10\times 0.05}}{0.05} = £7\,869.39.$$

Example 6.19 A cashflow is paid continuously from time 0 to time 10 years. The initial rate of payment is £1 000 p.a. and increases continuously thereafter at the rate 3% p.a. effective. Calculate the PV at time 0 of the cashflow, assuming a force of interest of 5% p.a.

Solution: The rate of payment is $\rho(t) = 1\,000(1.03)^t$ at time $t \in [0, 10]$. Then

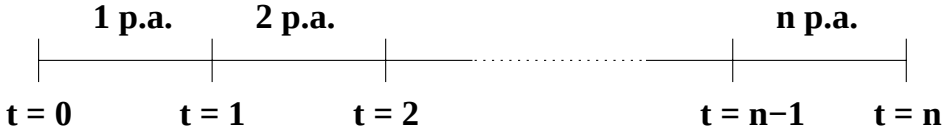
$$\begin{aligned} PV(0) &= \int_0^{10} \rho(t)e^{-\delta t}dt \\ &= \int_0^{10} 1\,000(1.03)^t e^{-0.05t}dt \\ &= 1\,000\int_0^{10} e^{t\ln(1.03)}e^{-0.05t}dt \\ &= 1\,000\int_0^{10} e^{-(0.05-\ln(1.03))t}dt \end{aligned}$$

Set $\hat{\delta} = 0.05 - \ln(1.03)$. Then

$$PV(0) = 1\,000\bar{a}^{\hat{\delta}}_{\overline{10}|} = 1\,000\frac{1 - e^{-10\times\hat{\delta}}}{\hat{\delta}} = £9\,044.16.$$

Increasing annuity: rate of payment k units p.a. in year k

Denote by $(I\bar{a})_{\overline{n}|}$ the PV at the start of a cashflow which is paid continuously for n years and has a rate of payment k units p.a. in year k, i.e.



Important formula:

$$(I\bar{a})_{\overline{n}|} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{\delta}.$$

Proof (one way):

$$\begin{aligned}(I\bar{a})_{\overline{n}|} &= \bar{a}_{\overline{1}|} + 2v\bar{a}_{\overline{1}|} + 3v^2\bar{a}_{\overline{1}|} + \cdots nv^{n-1}\bar{a}_{\overline{1}|} \\ &= \bar{a}_{\overline{1}|}\left(1 + 2v + 3v^2 + \cdots + nv^{n-1}\right) \\ &= \bar{a}_{\overline{1}|}(I\ddot{a})_{\overline{n}|} \\ &= \frac{1-v}{\delta}\frac{\ddot{a}_{\overline{n}|} - nv^n}{d} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{\delta}, \qquad \text{since } d = 1 - v.\end{aligned}$$

Increasing annuity: rate of payment t units p.a. at time t

Denote by $(\bar{I}\bar{a})_{\overline{n}|}$ the present value of a cashflow which is paid continuously for n years and has a rate of payment

$$\rho(t) = t, \qquad \text{at } t \in [0, n].$$

Important formula:

$$(\bar{I}\bar{a})_{\overline{n}|} = \frac{\bar{a}_{\overline{n}|} - nv^n}{\delta}.$$

Proof: For you to do as an exercise (see Tutorial 2).

Actuarial notation for accumulations to time n:

$$\begin{aligned}\bar{s}_{\overline{n}|} &= (1+i)^n\bar{a}_{\overline{n}|} \\ (I\bar{s})_{\overline{n}|} &= (1+i)^n(I\bar{a})_{\overline{n}|} \\ (\bar{I}\bar{s})_{\overline{n}|} &= (1+i)^n(\bar{I}\bar{a})_{\overline{n}|}.\end{aligned}$$

Example 6.20 Find the value at time 15 of a cashflow paying continuously at a rate $\rho(t)$ p.a. where

$$\rho(t) = \begin{cases} 10 & \text{for } 0 \le t < 5 \\ 12 & \text{for } 5 \le t < 15 \end{cases}$$

assuming a constant force of interest δ.

Solution: Can you see that

$$\text{PV}(15) = e^{15\delta} \times 10\bar{a}_{\overline{5}|} + e^{10\delta} \times 12\bar{a}_{\overline{10}|}$$

Alternatively, we can figure this out the long way:

$$\begin{aligned}
\text{PV}(15) &= \int_0^{15} \rho(t) e^{\int_t^{15} \delta(u)du} dt \\
&= \int_0^{15} \rho(t) e^{(15-t)\delta} dt \\
&= \int_0^{5} \rho(t) e^{(15-t)\delta} dt + \int_5^{15} \rho(t) e^{(15-t)\delta} dt \\
&= \int_0^{5} 10 e^{(15-t)\delta} dt + \int_5^{15} 12 e^{(15-t)\delta} dt \\
&= 10e^{15\delta} \int_0^{5} e^{-\delta t} dt + 12e^{15\delta} \int_5^{15} e^{-\delta t} dt \\
&= 10e^{15\delta} \bar{a}_{\overline{5}|} + 12e^{10\delta} \bar{a}_{\overline{10}|}.
\end{aligned}$$

Exercise 6.21 An investor buys an annuity which pays him cash continuously for 5 years. He receives payment in the first year of £3,000, thereafter his annual receipts increase by £500 per annum. Calculate the purchase price of the annuity, assuming that the annual effective rate of interest is 6%.

Exercise 6.22 An investor agrees to make investments continuously for the next 10 years. She decides that she can afford to invest $£20t$ at time t, $0 \leq t \leq 10$. Calculate the value of the investment at time 0, assuming that the annual effective rate of interest is 4%.

7 Duration, Convexity and Immunization

7.1 Duration

Consider a set of discrete future cash flows C_{t_k} to be paid at time t_k, $0 < t_1 < \ldots < t_n < T$ and a continuous cashflow at rate of payment $\rho(t)$ for $t \in [0, T]$.

The net present value of these cash flows at constant force of interest δ is

$$\text{NPV}(\delta) = \sum_{k=1}^{n} C_{t_k} e^{-\delta t_k} + \int_0^T \rho(t) e^{-\delta t} dt.$$

What happens if the force of interest suddenly changes - how will the NPV change? Will it increase or decrease it value? By how much will it increase or decrease? We introduce the concept of *duration* to estimate the latter. It measures the sensitivity of the NPV of a set of cashflows to changes in the interest rate.

Definition 7.1

The *(Macaulay) duration* $\tau(\delta)$ of these cashflows at the force of interest δ is

$$\begin{aligned}\tau(\delta) &= -\frac{1}{\text{NPV}(\delta)} \times \frac{d}{d\delta}\text{NPV}(\delta) \\ &= \frac{\sum_{k=1}^{n} t_k C_{t_k} e^{-\delta t_k} + \int_0^T t\rho(t) e^{-\delta t} dt}{\sum_{k=1}^{n} C_{t_k} e^{-\delta t_k} + \int_0^T \rho(t) e^{-\delta t} dt}.\end{aligned}$$

Remarks:

1. The duration is also called the *discounted mean term (DMT)*.
2. It is the weighted average time until payment of a set of cashflows, with the weights being the present value of each cashflow.

Use of duration

$\tau(\delta)$ measures the proportionate change in $\text{NPV}(\delta)$ due to a small change in the force of interest δ.

$$\tau(\delta) \approx -\frac{1}{\text{NPV}(\delta)} \frac{\Delta\text{NPV}(\delta)}{\Delta\delta} \qquad \Rightarrow \qquad \frac{\Delta\text{NPV}(\delta)}{\text{NPV}(\delta)} \approx -\Delta\delta \cdot \tau(\delta).$$

Thus, if the force of interest changes from δ_0 to δ_1 then:

$$\frac{\text{NPV}(\delta_1) - \text{NPV}(\delta_0)}{\text{NPV}(\delta_0)} \approx -(\delta_1 - \delta_0) \cdot \tau(\delta_0)$$

This can be a good approximation for *immediate small* changes in δ.

Aside: Duration calculated as

$$-\frac{1}{\text{NPV}(i)} \times \frac{d}{di}\text{NPV}(i)$$

is called the *modified duration*. It is not equal to the Macaulay duration; can you find the relationship between them? We consider only the Macaulay duration in this course.

Definition 7.2

An *n-year zero-coupon bond* is a bond which pays only a redemption amount in n years time and no coupons. If we buy £1 nominal of a n-year zero-coupon bond, this means that we receive £1 in n years' time.

Example 7.1 If £1 nominal of an 3-year zero-coupon bond costs £0.95, then an investor can pay £0.95 today to receive a payment of £1 in exactly 3 years' time.

Example 7.2 Suppose the force of interest is $\delta = 0.05$ p.a. How much should an investor pay today to buy £100 nominal of a 10-year zero-coupon bond?

Solution: $100e^{-10\delta} = 100e^{-10\times 0.05} = 100 \times 0.6065 = £60.65$.

Exercise 7.3 Show that the duration of an n-year zero-coupon bond (ZCB) is n years, i.e. it does not depend on the chosen force of interest.

Exercise 7.4

1. Show that the duration of an n-year annuity of £1 p.a. paid in arrears is

$$\frac{(Ia)_{\overline{n}|}}{a_{\overline{n}|}}.$$

2. Show that the duration of an n-year annuity of £1 p.a. paid continuously is

$$\frac{(\bar{I}\bar{a})_{\overline{n}|}}{\bar{a}_{\overline{n}|}}.$$

Example 7.5 Two investments providing the following two sets of cashflows

Time (years)	Cashflow 1	Cashflow 2
1	50	6.7
2	25	13.4
3	12	26.8
4	6	53.6

Calculate the NPV and the duration of the two sets of cashflows at $\delta = 0.05$.

$$\delta = 0.05 \implies v = e^{-0.05}.$$

$$\text{NPV}_1(\delta) = 85.4 = \text{NPV}_2(\delta)$$

$$\tau_1(\delta) = \frac{\{1 \times 50 \times v + 2 \times 25 \times v^2 + 3 \times 12 \times v^3 + 4 \times 6 \times v^4\}}{\text{NPV}_1(\delta)} = 1.68$$

$$\tau_2(\delta) = \frac{\{1 \times 6.7 \times v + 2 \times 13.4 \times v^2 + 3 \times 26.8 \times v^3 + 4 \times 53.6 \times v^4\}}{\text{NPV}_2(\delta)} = 3.22$$

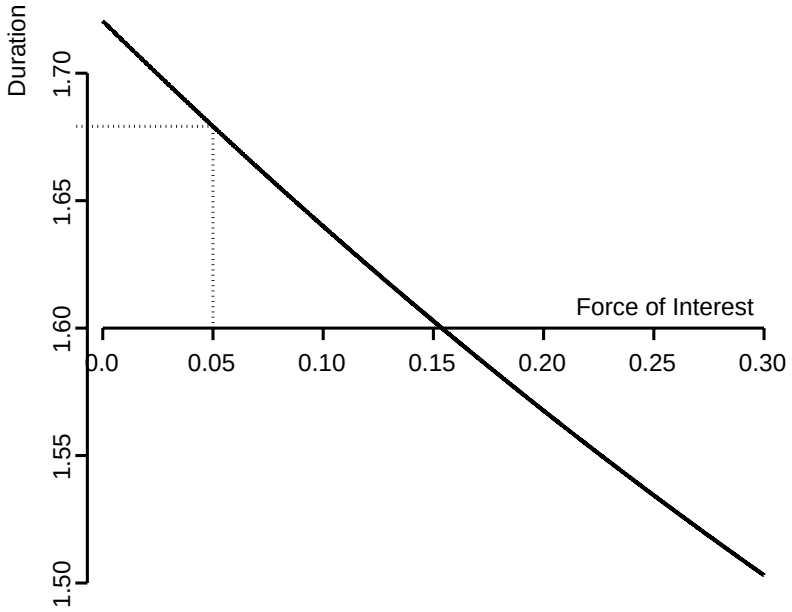


Figure 1: Duration of Cashflow 1 in Example 7.5.

Example 7.6 Two fixed-income securities:

A: term 10 years, coupon 12% p.a. paid annually in arrears, redemption at par.

B: term 25 years, coupon 3% p.a. paid annually in arrears, redemption 120%.

$$\begin{aligned}\tau_{\text{A}}(\delta) &= \frac{1 \times 12 \times v + 2 \times 12 \times v^2 + \cdots + 10 \times 12 \times v^{10} + 10 \times 100 \times v^{10}}{12a_{\overline{10}|} + 100v^{10}} \\ &= \frac{12\left(v + 2v^2 + \cdots + 10v^{10}\right) + 1000v^{10}}{12a_{\overline{10}|} + 100v^{10}} \\ &= \frac{12(Ia)_{\overline{10}|} + 1000v^{10}}{12a_{\overline{10}|} + 100v^{10}}.\end{aligned}$$

Similarly,

$$\tau_{\text{B}}(\delta) = \frac{3(Ia)_{\overline{25}|} + 120 \times 25 \times v^{25}}{3a_{\overline{25}|} + 120v^{25}}.$$

Assume the current force of interest is 4% p.a.

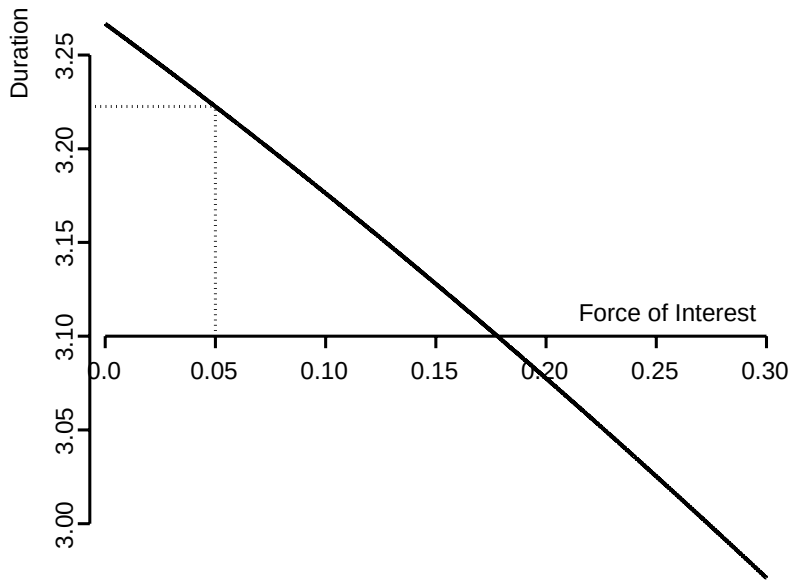


Figure 2: Duration of Cashflow 2 in Example 7.5.

δ	$\text{NPV}_{\text{A}}(\delta)$	$\text{NPV}_{\text{B}}(\delta)$	$\tau_{\text{A}}(\delta)$	$\tau_{\text{B}}(\delta)$
4%	163.97	90.61	7.1451	17.7975

In the following table, we compare the actual change in the NPV with the approximation of the change in the NPV using the duration, if the force of interest suddenly shifted from 4%.

Sudden shift in force of interest	Actual price change A	Actual price change B	Estimated price change $\approx -(\delta_1 - \delta_0)\tau(0.04)$ A	Estimated price change $\approx -(\delta_1 - \delta_0)\tau(0.04)$ B
From 4% to 3%	+7.465%	+19.92%	+7.1451%	+17.7975%
From 4% to 5%	-6.843%	-15.98%	-7.1451%	-17.7975%

For the actual price change, we used the following figures:

δ	$\text{NPV}_{\text{A}}(\delta)$	$\text{NPV}_{\text{B}}(\delta)$
3%	176.21	108.66
5%	152.75	76.13

7.2 Convexity

Definition 7.3

For any set of discrete cashflows $\{(C_{t_k}, t_k); k = 1, 2, \ldots, n\}$ and a continuous cashflow at rate of payment $\rho(t)$ for all $t \in [0, T]$, we define the *convexity* of the combined cashflows as

$$\begin{aligned}\mathrm{CONV}(\delta) &= \frac{1}{\mathrm{NPV}(\delta)} \frac{d}{d\delta^2} \mathrm{NPV}(\delta) \\ &= \frac{\sum_{k=1}^{n} t_k^2 C_{t_k} e^{-\delta t_k} + \int_0^T t^2 \rho(t) e^{-\delta t} dt}{\sum_{k=1}^{n} C_{t_k} e^{-\delta t_k} + \int_0^T \rho(t) e^{-\delta t} dt}.\end{aligned}$$

Exercise 7.7 Show that the convexity of an n-year zero-coupon bond (ZCB) is n^2 years, i.e. it does not depend on the chosen force of interest.

Exercise 7.8 1. Show that the convexity of an n-year annuity of £1 p.a. paid in arrears is

$$\frac{\sum_{k=1}^{n} k^2 v^k}{a_{\overline{n}|}}.$$

2. Show that the convexity of an n-year annuity of £1 p.a. paid continuously is

$$\frac{2(\bar{I}\bar{a})_{\overline{n}|} - n^2 v^n}{\delta \bar{a}_{\overline{n}|}}.$$

Exercise 7.9 By using a Taylor series expansion about $\delta = \delta_0$, show that

$$\mathrm{NPV}(\delta) \approx \mathrm{NPV}(\delta_0) \left(1 - (\delta - \delta_0)\tau(\delta_0) + \frac{(\delta - \delta_0)^2}{2} \mathrm{CONV}(\delta_0)\right).$$

Exercise 7.10 Convexity can be used to provide a better approximation of the change in NPV. In a spreadsheet, use the approximation in Exercise 7.9 to obtain better approximations for the change in NPV for the fixed-income securities A and B of Example 7.6. Compare the duration-convexity approximations to the duration-only approximations and the actual price changes.

7.3 Immunization

Set-up: financial institution (e.g. life insurance company or pension fund).

The institution has future net liability cashflows of L_{t_k} at time t_k, for $k = 1, 2, \ldots$.

It holds assets which provide cashflows A_{s_n} at time s_n, for $n = 1, 2, \ldots$.

Let the current force of interest be δ_0 p.a. We assume that

$$\mathrm{NPV}_{\mathrm{A}}(\delta_0) = \sum_{n=1}^{\infty} A_{s_n} e^{-\delta_0 s_n} = \mathrm{NPV}_{\mathrm{L}}(\delta_0) = \sum_{k=1}^{\infty} L_{t_k} e^{-\delta_0 t_k}. \tag{7.3}$$

Since the NPV is a function of the force of interest, equation (7.3) may no longer hold if the force of interest changes. Hence, the institution faces the following problem.

Problem: How to choose the assets so that an instantaneous change in the force of interest from δ_0 to δ_1 will not result in $\mathrm{NPV}_{\mathrm{A}}(\delta_1) < \mathrm{NPV}_{\mathrm{L}}(\delta_1)$?

"Perfect Matching": Choose $s_1 = t_1$, $s_2 = t_2$, ..., and then $A_{t_k} = L_{t_k}$ for all k (not usually a practical solution).

In cases in which "Perfect Matching" is not possible it might still be possible for the institution to choose there assets such that a small immediate change in the force of interest will not result in insufficient NPV of the assets. This provides at least some protection against the risk associated with a change in the interest rate.

The following theorem shows how a financial institution can protect itself against immediate small changes in the force of interest. This protection is called immunization.

Redington's theory of immunization

Theorem 7.11 *Suppose that the market force of interest is δ_0 p.a. If the following conditions hold:*

$$NPV_{A}(\delta_0) = NPV_{L}(\delta_0), \qquad \text{"NPV matching"} \tag{7.4}$$

$$\tau_{A}(\delta_0) = \tau_{L}(\delta_0), \qquad \text{"Duration matching"} \tag{7.5}$$

$$CONV_{A}(\delta_0) > CONV_{L}(\delta_0), \qquad \text{"Convexity condition"} \tag{7.6}$$

then there exists $\varepsilon > 0$ such that for all δ with $|\delta - \delta_0| < \varepsilon$ the following holds

$$NPV_{A}(\delta) \geq NPV_{L}(\delta),$$

with equality holding only at $\delta = \delta_0$.

Interpretation: If the conditions (7.4), (7.5) and (7.6) are fulfilled, then an *immediate, small* change in the force of interest from δ_0 to δ_1 will give:

$$\mathrm{NPV}_{\mathrm{A}}(\delta_1) > \mathrm{NPV}_{\mathrm{L}}(\delta_1).$$

Proof. Define $f(\delta) = \mathrm{NPV}_{\mathrm{A}}(\delta) - \mathrm{NPV}_{\mathrm{L}}(\delta)$.

$$\begin{aligned}
\text{Equation (7.4)} \quad &\Rightarrow \quad f(\delta_0) = 0. \\
\text{Equations (7.4) and (7.5)} \quad &\Rightarrow \quad NPV'_{\mathrm{A}}(\delta_0) - \mathrm{NPV}'_{\mathrm{L}}(\delta_0) = 0 \\
&\Rightarrow \quad f'(\delta_0) = 0 \\
&\Rightarrow \quad f \text{ has a turning point at } \delta_0. \\
\text{Equations (7.4) and (7.6)} \quad &\Rightarrow \quad NPV''_{\mathrm{A}}(\delta_0) - \mathrm{NPV}''_{\mathrm{L}}(\delta_0) > 0 \\
&\Rightarrow \quad f''(\delta_0) > 0 \\
&\Rightarrow \quad f \text{ has a minimum point at } \delta_0.
\end{aligned}$$

It follows that (i) $f(\delta)$ has a local minimum at $\delta = \delta_0$, and (ii) $f(\delta_0) = 0$.

Hence for all δ close to δ_0,

$$f(\delta) > f(\delta_0) = 0.$$

More precisely, there exists $\varepsilon > 0$ such that for all δ with $|\delta - \delta_0| < \varepsilon$, we have $\mathrm{NPV}_{\mathrm{A}}(\delta) \geq \mathrm{NPV}_{\mathrm{L}}(\delta)$,with equality holding only at $\delta = \delta_0$. □

Remarks:

1. If conditions (7.4),(7.5) and (7.6) hold, the financial institution is said be immunized (in the sense of Redington) against immediate small changes in the interest rate.
2. The financial institution is immunized against immediate small changes in the interest rate assuming a flat yield curve. (Generalizes to immediate parallel shifts of a yield curve.)
3. Suppose (7.4) and (7.5) hold and there are only discrete cashflows. Then (7.6) is equivalent to

$$\sum_{n=1}^{\infty}(s_n - \tau_{\mathrm{A}}(\delta_0))^2 A_{s_n} \exp(-\delta_0 s_n) > \sum_{k=1}^{\infty}(t_k - \tau_{\mathrm{L}}(\delta_0))^2 L_{t_k} \exp(-\delta_0 t_k) \tag{7.7}$$

   i.e. the "spread of the assets" around the common τ is greater than the "spread of the liabilities". A similar result holds for continuous cashflows.

Example 7.12 A life office has to pay £1 million in 10 years time. The effective rate of interest is $i = 4\%$ p.a.

The available securities are a 5-year and a 15-year zero-coupon bond (ZCB).

Calculate how much of each ZCB the life office should buy so that it is immunised in the sense of Redington.

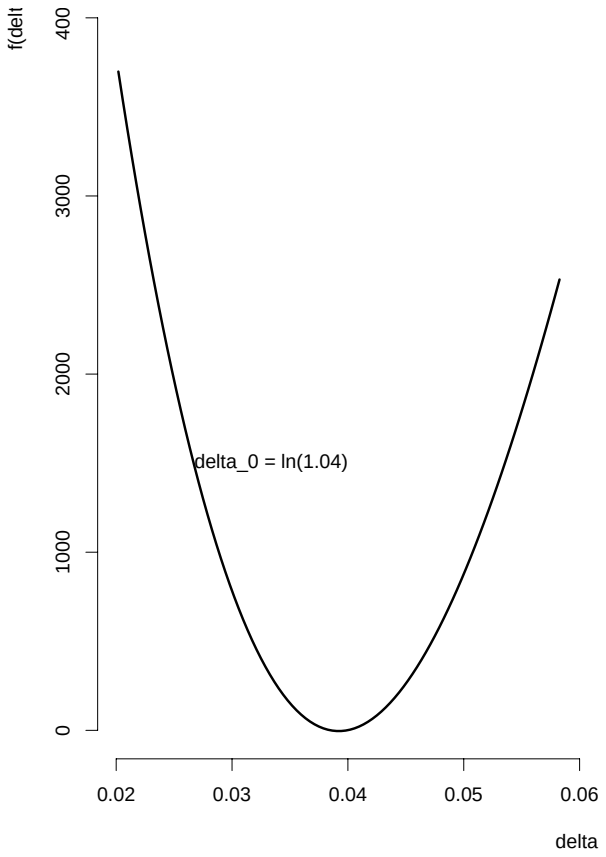


Figure 3: $\text{NPV}_A - \text{NPV}_L$ for i between 2% and 6% (i.e. $\delta \in [\ln 1.02, \ln 1.06]$), for Example 7.12.

Solution: Suppose £X is invested in the 5-year ZCB at $t = 0$ and £Y is invested in the 15-year ZCB at $t = 0$.

The value of the liability at time 0 is $\text{NPV}_{\text{L}} = £1\,000\,000 \times 1.04^{-10} = £675\,564$.

1. Equate NPVs: $\text{NPV}_{\text{A}} = \text{NPV}_{\text{L}} \Rightarrow X + Y = 675\,564$.

2. Equate durations:

$$\tau_{\text{A}} = \tau_{\text{L}} \qquad \Rightarrow \qquad \frac{5X + 15Y}{X + Y} = \frac{10\,(1\,000\,000)v^{10}}{1\,000\,000v^{10}}.$$

   Hence we have two linear equations in two unknowns:

$$X + Y = 675\,564$$
$$5X + 15Y = 10.$$

   Solve them for X and Y to find $X = £337\,782$ and $Y = £337\,782$, i.e. half the assets are invested in the 5-year ZCB and half are in the 15-year ZCB at $t = 0$.

3. Check if the convexity condition (7.6) holds.

$$\text{CONV}_A = \frac{5^2X + 15^2Y}{X + Y} = 125 > 100 = \frac{10^2\,(1\,000\,000)v^{10}}{1\,000\,000v^{10}} = \text{CONV}_L.$$

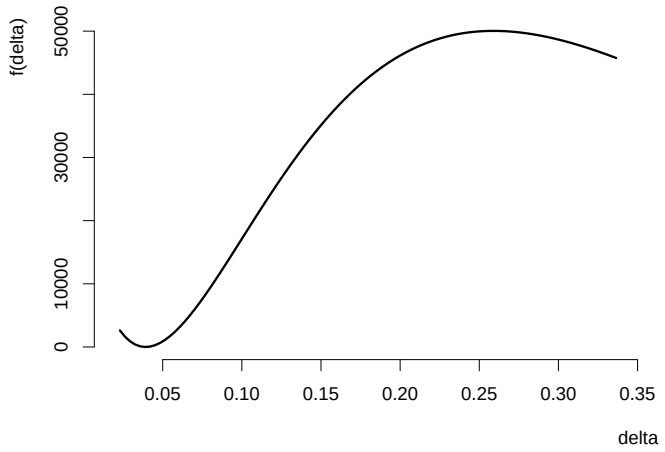


Figure 4: $\text{NPV}_A - \text{NPV}_L$ for i between 0% and 40%, for Example 7.12.

Since it holds, the financial institution is immunized.

Remarks:

1. Price of the 5-year ZCB per £1 nominal is $P_5 = v^5 = 1.04^{-5} = 0.82193$. Investing $X = £337\,782$ at time 0 in the 5-year ZCB means that the insurer purchases

$$\frac{X}{P_5} = \frac{337\,782}{0.82193} = £410\,962 \text{ nominal of the 5-year ZCB.}$$

2. Price of the 15-year ZCB per £1 nominal is $P_{15} = v^{15} = 1.04^{-15} = 0.55526$. Investing $Y = £337\,782$ at time 0 in the 15-year ZCB means that the insurer purchases

$$\frac{Y}{P_{15}} = \frac{337\,782}{0.55526} = £608\,331 \text{ nominal of the 15-year ZCB.}$$

Exercise 7.13 Show that with the solution above we have:

1. If interest rates move to 3% p.a. effective, NPV(3%) of (assets-liabilities) $\approx$ +£868.
2. If interest rates move to 5% p.a. effective, NPV(5%) of (assets-liabilities) $\approx$ +£703.

Hint: Use the nominal amounts of the ZCBs to calculate their prices under the new interest rate.

Exercise 7.14 Show that if all money would have been invested in the 5-year ZCB:

1. If interest rates move to 3% p.a. effective, NPV(3%) of (assets-liabilities) $\approx$ -£35 093.
2. If interest rates move to 5% p.a. effective, NPV(5%) of (assets-liabilities) $\approx$ +£30 088.

Hence we see that immunization works against profits as well as losses.

Exercise 7.15 An insurance company has issued a number of policies for which a total liability of £1 000 000 is payable in exactly 10 years' time. The insurance company holds sufficient funds

to cover the liability on the basis of a force of interest of δ per annum and intends to invest this money solely in the purchase of a 5-year and a 20-year zero-coupon bond (ZCB).

What proportion of the company's funds should be invested in the 5-year ZCB to ensure that the discounted mean term of the assets equals the discounted mean term of the liabilities at a constant force of interest of δ per annum given that this is also the market basis for pricing all stocks?

(Answer: 2/3)

8 Arbitrage and Forward Contracts

We will now consider securities that have random (uncertain) future prices. Trading in these securities yields random future cash flows.

8.1 Arbitrage and the No-Arbitrage Assumption

The concept of arbitrage is one of the most important concepts in modern financial mathematics. Loosely speaking, an arbitrage is a risk-free trading profit.

More formally we say that an arbitrage opportunity exists in a financial market if either:

1. an investor can make a deal which would give an immediate profit with no risk of future loss, or
2. an investor can make a deal at no initial cost, has no risk of future loss and a non-zero probability of future profit.

Example 8.1
- Bank A charges interest at 5% p.a. effective on loans.
- Bank B provides interest at 6% p.a. effective on deposits.

There is an arbitrage opportunity! e.g. borrow as much as possible from A, deposit it with B, repay A eventually and make a profit.

No initial costs, but non-zero probability (here, probability 1) of making a profit.

Pricing of financial instruments often involves *short-selling*. Short-selling means selling a security that is not owned with the intention of buying it later. Short-selling yields profits when the price of the security goes down and a loss if the price goes up.

The mechanics of short-selling

Your broker sells on your behalf shares of a security which are owned by another client. All income of that security during the period of lending has to be given to the owner. Shares must be bought back and given back to the owner. Margins are required for security reasons.

We illustrate the idea of short-selling with an example. Denote by S_t the price of an IBM stock at time t. A dividend of £0.50 per unit IBM stock is payable at time $t_1 = 3/12$. We assume a constant force of interest of $\delta = 4\%$ p.a. and we ignore any borrowing fees in this example.

1. *At* $t = 0$, suppose $S_0 = £10$. A short-seller borrows one unit of IBM stock from a broker and immediately sells it in the market for £10. The short-seller deposits the cash in the bank.

   Short-seller's position at $t = 0$:

   - owes one unit of IBM stock to the broker, and

- £10 in the bank, accumulating at the force of interest $\delta = 4\%$ p.a.

2. *At* $t = t_1 = 3/12$, the dividend of £0.50 per unit stock is paid by IBM. Therefore, the short-seller must pay £0.50 to the broker.

   Short-seller's position at $t = t_1 = 3/12$:

   - owes one unit of IBM stock to the broker, and
   - $£(10e^{\delta t_1} - 0.50)$ in the bank, accumulating at the force of interest $\delta = 4\%$ p.a.

3. *At* $t = T = 9/12$, the short-seller decides to "close out" his/her position. Suppose that $S_T = £9$. The short-seller buys one unit of IBM stock from the market at the market price $S_T = £9$ and passes the stock certificate to the broker.

   Short-seller's position at $t = T = 9/12$:

   - $£((10e^{\delta t_1} - 0.50)e^{\delta(T-t_1)} - 9)$ in the bank.

Example 8.2 An "arbitrage opportunity" involving short-selling
Two assets traded over 1 time period on a stock exchange.
Suppose that during this period market prices either go up or down.

		Price at time 1	
Asset	Price now	Market falls	Market rises
1	7	4	8
2	9	6	12

Trading strategy:

At time 0:

- Sell 3 units of asset 1
- Buy 2 units of asset 2

$\Rightarrow$ income (at time 0):

$$\underbrace{3 \times 7}_{\text{"(short-)selling asset 1"}} - \underbrace{2 \times 9}_{\text{"buying asset 2"}} = 21 - 18 = 3$$

At time 1:

- Buy 3 units of asset 1
- Sell 2 units of asset 2

$\Rightarrow$ income (at time 1) if market falls:

$$\underbrace{2 \times 6}_{\text{"selling asset 2"}} - \underbrace{3 \times 4}_{\text{"buying asset 1"}} = 0$$

$\Rightarrow$ income (at time 1) if market rises:

$$\underbrace{2 \times 12}_{\text{"selling asset 2"}} \quad - \quad \underbrace{3 \times 8}_{\text{"buying asset 1"}} \quad = 0$$

$\Rightarrow$ Arbitrage

No arbitrage, if at time 0:

$$\text{"price of asset 1"} = \tfrac{2}{3} \times \text{"price of asset 2"}$$

No Arbitrage Assumption

No Arbitrage Assumption: No arbitrage opportunity can exist.

Remarks

1. Reasonable assumption in liquid markets. Trades of arbitrageurs will eliminate arbitrage opportunities (by force of supply and demand).
2. No arbitrage implies that if two sets of assets have exactly the same payout (i.e. the same value at some future time), they must have the same value now.

   e.g. in Example 8.2, 3 $\times$ asset 1 = 2 $\times$ asset 2 at time 1. Hence at time 0: price of asset 1 = $\frac{2}{3}$ price of asset 2.
3. The *replicating portfolio* for a given set of assets is a portfolio with the same value at some future time of the assets.

   No arbitrage implies that a replicating portfolio must have the same value as the given set of assets at any intermediate time.
4. No arbitrage and replicating portfolio are key ideas in modern finance.

Exercise 8.3 A company is to receive £100 at the end of each of the next 2 years. In the market, there is a 1-year zero-coupon bond (ZCB) with price £0.97 per £1 nominal and a 2-year ZCB with price £0.94 per £1 nominal.

1. Find a replicating portfolio for the company's future cashflows.
2. Calculate the no-arbitrage price [i.e. the market-consistent price] of the company's future cashflows.

Exercise 8.4 Consider a market in which there is only:

- 3-year fixed-interest bond which is redeemable at par and has an annual coupon of 5% p.a. payable annually in arrears, and
- n-year ZCB with price £P_n per £1 nominal for $n = 1, 2, 3$.

1. Find a replicating portfolio for £100 nominal of the fixed-interest bond.
2. If $P_1 = 0.98$, $P_2 = 0.96$ and $P_3 = 0.94$, calculate the no-arbitrage price per £100 nominal of the fixed-interest bond.

Exercise 8.5 A company projects its future cashflows as follows:

Time t (years)	0	1	2	3	4	5
Cashflow at t (£m)	0	10	20	10	10	110

In the market, there is

- 5-year fixed-interest bond which is redeemable at par and has an annual coupon of 10% p.a. payable annually in arrears, and
- 2-year ZCB.

1. Find a replicating portfolio for the cashflows of the company.
2. If the 5-year FI bond sells for £127 per £100 nominal and the 2-year ZCB sells for £0.92 per £1 nominal, then calculate the no-arbitrage price [i.e. the market-consistent price] of the company's cashflows.

Exercise 8.6 An insurance company has a liability to pay £100 in exactly 2 years. In the market, there is a 2-year ZCB with price £0.94 per £1 nominal.

1. Find a replicating portfolio for the insurance company's liability.
2. Calculate the no-arbitrage price [i.e. the market-consistent price] of the insurance company's liability.

Exercise 8.7 A company projects its future cashflows as follows:

Time t (years)	0	1	2	3	4	5
Cashflow at t (£m)	0	10	20	10	10	80

In the market, there is

- 5-year fixed-interest bond which is redeemable at par and has an annual coupon of 10% p.a. payable annually in arrears,
- 2-year ZCB, and
- 5-year ZCB.

1. Find a replicating portfolio for the cashflows of the company.
2. If 5-year FI bond sells for £127 per £100 nominal, the 2-year ZCB sells for £0.92 per £1 nominal and the 5-year ZCB sells for £0.82 per £1 nominal, then calculate the no-arbitrage price [i.e. the market-consistent price] of the company's cashflows.

Exercise 8.8 For Exercise 8.4,

1. If the market price of the fixed-interest bond was £106 per £100 nominal, explain how to make an arbitrage profit.
2. If the market price of the fixed-interest bond was £110 per £100 nominal, explain how to make an arbitrage profit.

[Hint: buy low, (short-)sell high]

Exercise 8.9 Consider the following portfolio,

- 1 unit of a non-dividend paying IBM stock, and
- short position in a forward contract to sell 1 unit of IBM stock in 5 years' time for £K to a counterparty.

Denote by S_t the price of the stock at time t years. The current time is $t = 0$ and $S_0 = £185$. In the market, there is a 5-year ZCB selling for £0.82193 per £1 nominal.

1. Calculate the current price of the portfolio.
2. What are the future cashflows from the portfolio?
3. Find a replicating portfolio for the portfolio.
4. Assuming no-arbitrage, find K, i.e. find K by equating the current price of the portfolio with the current price of its replicating portfolio.
5. Suppose there was no 5-year ZCB in the market. Instead, there is a bank account paying interest at 4% p.a. effective. Would the value of K calculated in part (iv) change? [Hint: consider if the price of the 5-year ZCB is consistent with the interest rate at the bank].

Solutions to selected exercises.

8.3 (i) { £100 nominal of 1-year ZCB, £100 nominal of 2-year ZCB }; 8.3 (ii) £191;

8.4 (i) { £5 nominal of 1-year ZCB, £5 nominal of 2-year ZCB, £105 nominal of 3-year ZCB }; 8.4 (ii) £108.40;

8.5 (i) {£100 million nominal of 5-year FI bond, £10 million nominal of 2-year ZCB }; 8.5 (ii) £136.2 million;

8.6 (i) { -£100 nominal of 2-year ZCB }, i.e. short-sell 2-year ZCB; 8.6 (ii) £94;

8.7 (i) { £100 million nominal of 5-year FI bond, £10 million nominal of 2-year ZCB, -£30 million nominal of 5-year ZCB [i.e. short-sell 5-year ZCB]}; 8.7 (ii) £111.6 million;

8.8 (i) Portfolio = { Buy £100 nominal of FI bond at £106, short-sell the replicating portfolio to get £108.40 }, Arbitrage profit = £(108.40 - 106) = £2.40; 8.8 (ii) Portfolio = { Short-sell £100 nominal of FI bond to get £110, buy the replicating portfolio at £108.40 }, Arbitrage profit = £(110-108.40) = £1.60;

8.9 (i) $S_0 = £185$; 8.9 (ii) Cashflow of: £K at $t = 5$; 8.9 (iii) £K nominal of the 5-year ZCB; 8.9

(iv) £225.08; 8.9 (v) No, since $1.04^{-5} = 0.82193$;

8.2 Forward Price and Hedging

Set-up: Asset: current price is known, future price is uncertain (could be an (equity) share, fixed amount of foreign currency,...).

Example 8.10 A British company knows that it will receive USD 1 million in 3 months.

Problem: Unknown future exchange rate!

Company contacts bank and agrees to sell USD 1 million in 3 months' time for an agreed exchange rate £0.6854 per US dollar.

Definition 8.1

A *forward contract* is an agreement to buy (or sell respectively) an asset at a certain future time (called the *delivery time*) for a certain price.
The price agreed at time $t = 0$ to be paid at delivery time $t = T$ is called the *delivery price*.
It is chosen such that the value of the forward contract is zero at time 0.

We emphasis that there is no payment made at time 0 when the agreement is made. The payment is made at the delivery time T.

time 0	time T
Forward contract struck, no payments made	Specified asset supplied and payments made

Definition 8.2

The no-arbitrage *forward price* for a particular forward contract at a specified time $t \geq 0$ is the delivery price that would make the forward contract have zero value at that time t.

Example 8.11 (Example 8.10 continued) One month later, the delivery price remains £0.6854 per US dollar, but the forward price will be different (in general).

General Terminology

The following terminology is used in the industry.

Long position in forward contract = agree to buy the asset

Short position in forward contract = agree to sell the asset

Risk-free rate of interest = rate of interest that can be earned without assuming any risks (e.g. yield on Treasury bills).

Question: How to calculate the forward price of an asset? We will use the replicating portfolio idea and then apply the no-arbitrage assumption to calculate the forward price of an asset.

Assumptions:

- no arbitrage,
- no transaction costs, and
- there is a known risk-free force of interest δ at which money can be invested and borrowed.

Notation:

T is the delivery time (in years)

S_t is the price of the asset at time $t \in [0, T]$

K is the forward price.

Constructing replicating portfolios

In the market there are two assets:

- a bank acount from which any investor can borrow money from or deposit money at a constant force of interest δ p.a.
- a stock with price S_t at time t.

We define

$$\begin{aligned} \textit{Portfolio } A = \{ &\text{ long position in the forward contract struck at time 0 to exchange} \\ &1 \text{ unit of the stock at time } T \text{ for price } £K \ \} \end{aligned}$$

An investor enters a forward contract at time 0 to buy 1 unit of the stock at time T for price $£K$, i.e. the investor has bought Portfolio A at time 0. Note that the counterparty to the forward contract, i.e. the short position, has short-sold Portfolio A.

The value of the contract to the investor at time 0 is zero. The value of the contract to the investor at time T is $-K + S_T$, corresponding to the investor paying $£K$ for one unit of the stock at time T, which can be sold in the market at price S_T at time T.

Our goal is to find K. We do this by

(Step 1) constructing the replicating portfolio, i.e. constructing a portfolio which replicates the value of Portfolio A at time T.

(Step 2) applying the no-arbitrage assumption to say that the value of Portfolio A must equal the value of its replicating portfolio at time 0. By equating the time 0 values, we find K.

Non-dividend-paying stock

Suppose that the stock pays no dividends.

Step 1. Construct the replicating portfolio.

At time 0,

$$\textit{Replicating portfolio} = \{\ x \text{ units of cash, and} \\ y \text{ units of the stock } \}$$

We want to determine x and y.

The value of the replicating portfolio at time T is $xe^{\delta T} + yS_T$. For this time T-value to replicate the time T-value of Portfolio A, we must have

$$xe^{\delta T} + yS_T = -K + S_T \qquad \Rightarrow \qquad y = 1 \quad \text{and} \quad x = -Ke^{-\delta T}.$$

Note that the value of the replicating portfolio at time 0 is $x + yS_0$.

$$x + yS_0 = -Ke^{-\delta T} + S_0.$$

Step 2. Apply the no-arbitrage assumption.

By no-arbitrage, the time 0 value of Portfolio A must equal the time 0 value of its replicating portfolio, i.e.

$$0 = -Ke^{-\delta T} + S_0 \qquad \Rightarrow \qquad K = S_0 e^{\delta T}.$$

Cash dividend-paying stock

Suppose that the stock pays a known cash dividend of £C per unit of stock held at known time $u \in (0, T]$.

Step 1. Construct the replicating portfolio.

At time 0,

$$\textit{Replicating portfolio} = \{\ x \text{ units of cash, and} \\ y \text{ units of the stock } \}$$

We want to determine x and y. However, we must take care to account for the dividend payment.

The value of the replicating portfolio at time u_- (i.e. just before the dividend is paid) is $xe^{\delta u_-} + yS_{u_-}$.

The value of the replicating portfolio at time u (i.e. just after the dividend is paid) is $(xe^{\delta u} + yC) + yS_u$, i.e. at time u,

$$\textit{Replicating portfolio} = \{\ xe^{\delta u} + yC \text{ units of cash, and} \\ y \text{ units of the stock } \}$$

The value of the replicating portfolio at time T is

$$(xe^{\delta u} + yC)e^{\delta(T-u)} + yS_T = xe^{\delta T} + yCe^{\delta(T-u)} + yS_T.$$

For this time T-value to replicate the time T-value of Portfolio A, we must have

$$xe^{\delta T} + yCe^{\delta(T-u)} + yS_T = -K + S_T \qquad \Rightarrow \qquad y = 1 \quad \text{and} \quad x = -(K + Ce^{\delta(T-u)})e^{-\delta T}.$$

Note that the value of the replicating portfolio at time 0 is $x + yS_0$.

$$x + yS_0 = -(K + Ce^{\delta(T-u)})e^{-\delta T} + S_0.$$

Step 2. Apply the no-arbitrage assumption.

By no-arbitrage, the time 0 value of Portfolio A must equal the time 0 value of its replicating portfolio, i.e.

$$0 = -(K + Ce^{\delta(T-u)})e^{-\delta T} + S_0 \qquad \Rightarrow \qquad K = (S_0 - Ce^{-\delta u})e^{\delta T}.$$

Exercise 8.12

Suppose the stock pays a known cash dividend of £C_k per unit of stock held at known time $u_k \in (0, T]$, for $k = 1, 2$. Show that if no-arbitrage holds, the forward price for Portfolio A is

$$K = (S_0 - C_1e^{-\delta u_1} - C_2e^{-\delta u_2})e^{\delta T}.$$

Exercise 8.13 Consider

Portfolio $B = \{$ short position in the forward contract struck at time 0 to exchange 1 unit of the stock at time T for price £K $\}$

1. Find the no-arbitrage forward price K assuming:
   a) the stock pays no dividends.
   b) the stock pays a known cash dividend of £C per unit of stock held at known time $u \in (0, T]$,
2. How does Portfolio B relate to Portfolio A? How does the replicating portfolio for Portfolio B relate to the replicating portfolio for Portfolio A? You should find that buying B is equivalent to selling A, and vice versa. Similarly for the replicating portfolios.

Exercise 8.14 Suppose that the stock pays no dividends and the assets in the market are

- a bank acount from which any investor can borrow money from or deposit money at a force of interest $\delta(t)$ p.a, with
$$\delta(t) = \begin{cases} \delta_1 & \text{for } t \in [0, u) \\ \delta_2 & \text{for } t \in [u, T]. \end{cases}$$
- a stock with price S_t at time t.

Show that if no-arbitrage holds, the forward price for Portfolio A is

$$K = S_0e^{\delta_1 u + \delta_2(T-u)}.$$

Exercise 8.15 Suppose that the stock pays no dividends and the assets in the market are

- a T-year zero-coupon bond with price P_t at time t per £1 nominal.
- a stock with price S_t at time t.

Show that if no-arbitrage holds, the forward price for Portfolio A is

$$K = \frac{S_0}{P_0}.$$

Hint: the replicating portfolio at time 0 consists of £x nominal of the T-year zero-coupon bond and y units of the stock.

Example 8.16 Consider a forward contract in which the parties agree to exchange 1 unit of a foreign currency (e.g. 1 USD) at time T for forward price £K.

Let S_t denote the price in £ of 1 unit of foreign currency. For example, $S_0 = £0.62716$ per 1 USD, which means USD5 costs $5S_0 = £3.1358$.

Denote by δ_f the risk-free force of interest in foreign currency. Show that the no-arbitrage forward price for a T-year forward contract is

$$K = S_0 \exp\left\{(\delta - \delta_f)T\right\}.$$

Solution The replicating portfolio at time 0 consists of £x of the domestic currency and y units of the foreign currency. Thus the value of the replicating portfolio at time 0 is £$(x + yS_0)$.

At time T, the value of the replicating portfolio is $xe^{\delta T} + ye^{\delta_f T} S_T$.

As the net cashflow from the forward contract is $-K + S_T$, we set

$$xe^{\delta T} + ye^{\delta_f T} S_T = -K + S_T$$

and solve to find $x = -Ke^{-\delta T}$ and $y = e^{-\delta_f T}$. Applying no-arbitrage, we find that $K = S_0 \exp\left\{(\delta - \delta_f)T\right\}$.

Hedging

Consider the investor who has agreed to supply the asset at time T for the forward price $K = S_0 e^{\delta T}$.

Strategy 1: Do nothing before time T. Buy the asset at time T and deliver if for the price K.

Receive:	at time $t = 0$:	0
	at time $t = T$:	$K - S_T$ (could be negative)

Strategy 2: Borrow $Ke^{-\delta T}$ $(= S_0)$ at the risk-free force of interest δ, to be repaid at time T, and buy the asset now. Receive K at time T and repay the loan.

Receive:	at time $t = 0$:	0
	at time $t = T$:	0

Remarks:

- Strategy 2 $\Rightarrow$ no chance of profit and no risk of loss.
- Strategy 2 is called a *static hedge* for the forward contract.
- In general a *hedge portfolio* is a portfolio which eliminates or reduces the risk associated with an investment.
- There is a relationship between pricing and hedging.

Arbitrage strategies

We assume that the stock pays no dividends. However, exactly the same argument applies no matter which portfolio or type of stock we consider.

Consider

Portfolio $A' = \{$ long position in the forward contract struck at time 0 to exchange 1 unit of the stock at time T for price £K' $\}$

An arbitrage strategy can be found if the forward price K' agreed for Portfolio A' is not equal to the no-arbitrage forward price $K = S_0 e^{\delta T}$.

Consider the replicating portfolio for A'. It can be easily shown that the time 0 value of the replicating portfolio is $-K'e^{-\delta T} + S_0$. Substituting $S_0 = Ke^{-\delta T}$, we find

$$\begin{aligned}\text{Time 0 value of replicating portfolio } &= -K'e^{-\delta T} + S_0 = -K'e^{-\delta T} + Ke^{-\delta T} \\ &= (-K' + K)e^{-\delta T}.\end{aligned}$$

- If $K' > K$ then the time 0 value of the replicating portfolio is less than zero. In that case, an arbitrage strategy is to

  At time 0, buy the replicating portfolio and sell Portfolio A',

  resulting in an immediate profit of $-(-K' + K)e^{-\delta T}$ with no risk of future loss.

- If $K' < K$ then the time 0 value of the replicating portfolio is greater than zero. In that case, an arbitrage strategy is to

  At time 0, sell the replicating portfolio and buy Portfolio A',

  resulting in an immediate profit of $(-K' + K)e^{-\delta T}$ with no risk of future loss.

Example 8.17 (Non-dividend paying stock) The risk-free force of interest is 5% p.a. and the stock price is $S_0 = 30$ today. The no-arbitrage forward price for a 2 year forward contract is $K = S_0 e^{2\delta} = 33.16$.

1. Assume that the agreed forward price to exchange 1 unit of a non-divided paying stock in 2 years' time is K'=£35.

   An arbitrageur can adopt the following strategy:

		Today (time t=0)	In 2 years (time $T = 2$)
1.	Borrow £30 at $\delta = 0.05$ for 2 years	+30	$-30e^{2\times 0.05} = -33.16$
2.	Buy the stock	-30	0
3.	Enter a f.c. to sell 1 stock for £35 (=K') in 2 years	0	+35
		0	$35 - 33.16 = 1.84$

2. Assume that the agreed forward price to exchange 1 unit of a non-divided paying stock in 2 years' time is K''=£31.

   An arbitrageur can adopt the following strategy:

		Cashflow today (time t=0)	Cashflow in 2 years (time $T = 2$)
2.	Short the stock	+30	0
1.	Lend £30 at $\delta = 0.05$ for 2 years	−30	$+30e^{2\times 0.05} = 33.16$
3.	Enter a f.c. to buy 1 stock for £31 (=K'') in 2 years	0	-31
		0	$33.16 - 31 = 2.16$

Example 8.18 (Currencies) Agree to buy 1 unit of a foreign currency (e.g. 1 USD) at time T for forward price $£K$.

Let S_t denote the price in £of 1 unit of foreign currency. Denote by δ_f the risk-free force of interest in foreign currency. Recall from Example 8.16 that the no-arbitrage forward price for a T-year forward contract is $K = S_0 \exp\{(\delta - \delta_f)T\}$.

Assume that the agreed forward price to exchange 1 unit of foreign currency in 2 years' time is $K' > S_0 \exp\{(\delta - \delta_f)T\}$

One possible arbitrage strategy is then:

	Cashflow today (time t=0)	Cashflow in 2 years (time $T = 2$)
Borrow $S_0 \exp(-\delta_f T)$ in domestic currency at rate δ p.a. for T years	$S_0 \exp(-\delta_f T)$	$-S_0 \exp(-\delta_f T) \exp(\delta T)$
Buy $\exp(-\delta_f T)$ of the foreign currency and invest at rate δ_f p.a. for T years	$-S_0 \exp(-\delta_f T)$	
Enter a f.c. to sell 1 unit of foreign currency at T for price K'	0	K'
	0	$K' - S_0 \exp((\delta - \delta_f)T)$

8.3 The Value of a Forward Contract

Notation: f_t = value of a forward contract at time t, $t \in [0, T]$

Remark:

- $f_0 = 0$
- $f_T = \begin{cases} S_T - K & \text{(long position)} \\ K - S_T & \text{(short position)} \end{cases}$

Long Forward Contract Consider the following two portfolios at time t.

Portfolio A: Hold the existing long forward contract which was struck at time 0 with delivery price K_0.

Value of portfolio A at time t is f_t.

Portfolio B: Enter a new forward contract to buy the asset for delivery price K_t at time T. Deposit $(K_t - K_0)e^{-\delta(T-t)}$ in the bank account.

Value of portfolio B at time t is $0 + (K_t - K_0)e^{-\delta(T-t)}$.

At time T,

- value of portfolio A: $S_T - K_0$.
- value of portfolio B: $S_T - K_t + (K_t - K_0)e^{-\delta(T-t)}e^{\delta(T-t)} = S_T - K_0$.

Thus portfolio B replicates portfolio A.

By no arbitrage, { Value of portfolio A at time t } = {Value of portfolio B at time t } and hence

$$f_t = (K_t - K_0)e^{-\delta(T-t)}.$$

e.g. $f_{0.5} = 0.83$ means that Portfolio A is worth £0.83 at time 0.5. If the forward contract is

to be terminated at time 0.5, then the long position in the forward contract struck at time 0 (i.e. the buyer of Portfolio A) requires a payment of £0.83 from the short position.

e.g. $f_1 = -1.25$ means that Portfolio A is worth - £1.25 at time 1. If the forward contract is to be terminated at time 1, then the long position in the forward contract struck at time 0 (i.e. the buyer of Portfolio A) will have to pay £1.25 to the short position.

Short Forward Contract Using a similar argument, we find

$$f_t = (K_0 - K_t)\exp(-\delta(T-t)).$$

Example 8.19 3-month forward contract on USD.
Current exchange rate: USD 1 = GBP 0.535.
3-month UK risk-free interest rate: 4.72% p.a. $\Rightarrow$ $\delta = \log(1.0472)$.
3-month US risk-free interest rate: 2.4% p.a. $\Rightarrow$ $\delta_f = \log(1.024)$.

Forward price at $t = 0$, with $T = 3/12$,

$$\begin{aligned} K_0 &= S_0 \exp\{(\delta - \delta_f)T\} \\ &= 0.535 \exp\left\{(\log(1.0472) - \log(1.024))\,\frac{3}{12}\right\} \\ &= 0.5380049 \end{aligned}$$

One month later:

Exchange rate: USD 1 = GBP 0.52.
2-month UK risk-free interest rate: 4.82% p.a. $\Rightarrow$ $\delta = \log(1.0482)$.
2-month US risk-free interest rate: 2.56% p.a. $\Rightarrow$ $\delta_f = \log(1.0256)$.
Forward price at $t = 1/12$ is

$$\begin{aligned} K_{1/12} &= S_{1/12} \exp\{(\delta - \delta_f)(T - 1/12)\} \\ &= 0.52 \exp\left\{(\log(1.0482) - \log(1.0256))\,\frac{2}{12}\right\} \\ &= 0.5218925 \end{aligned}$$

Value of short forward contract:

$$\begin{aligned} f_{1/12} &= (K_0 - K_{1/12})\exp(-\delta(T-t)) \\ &= (0.5380049 - 0.5218925)\exp\left(-\log(1.0482) \times \frac{2}{12}\right) = 0.01598648 \end{aligned}$$

9 Term Structure of Interest Rates

9.1 Yield Curves

- The *term structure of interest rates* is the variation of bond yields with the terms to redemptions of the bonds.
- Graphically, this is also called a *yield curve*.
- Yields are calculated from market prices. They are determined in the market by demand and supply.
- Yield curves vary from day to day. They can be
  - increasing (upward-sloping), sometimes called "normal",
  - decreasing (downward-sloping), sometimes called "inverted",
  - flat (all yields are equal), or
  - inconsistent term to redemption/yield relationship, sometimes called "humped" (e.g. short-term and long-term yields are equal, but medium-term yields vary).
- Flat and humped yield curves indicate uncertainty. As certainty in the market increases, the yield curve moves towards a normal or an inverted shape.
- There are 3 observed characteristics of the term structure of interest rates:
  - The change in yields of different term bonds tends to move in the same direction.
  - The yields on short-term bonds are more volatile than long-term bonds.
  - The yields on long-term bonds are often higher than short-term bonds.
- Why does the yield curve have the shape it does?

Explanations for the shape of the yield curve

The main explanation is the Expectations Theory. It explains

- The change in yields of different term bonds tends to move in the same direction.
- The yields on short-term bonds are more volatile than long-term bonds.

The third observation:

- The yields on long-term bonds are often higher than short-term bonds.

can be explained by the Liquidity Preference Theory.

An additional explanation is the Market Segmentation Theory/Preferred Habitat Theory, which can be used in conjunction with the Expectations Theory and the Liquidity Preference Theory to

explain the shape of the yield curve.

Expectations Theory

- Long-term interest rates reflect investors' expectations of short-term interest rates in the future.
- This means that the long-term rate is composed of all the short term rates up to that long term.
- i.e. the long-term rate up to time T is the geometric mean of all the intervening short-term rates up to time T.

Liquidity Preference Theory

- The more risky the investment, the higher the *expected* return is required to be in order to entice investors.
- Long-dated bonds are more risky than short-dated bonds: when interest rates move, the price of a long-dated bond changes more than the price of a short-dated bond.
- Therefore, investors expect a higher interest rate on long-dated bonds because of the extra risk.

Together, the Expectations Theory and Liquidity Preference Theory suggest that an upward-sloping yield curve is more common than a downward-sloping one.

This is observed in practice.

Market Segmentation Theory

- Different investors are attracted to different types of bonds.
- This can be due to investment requirements,
- e.g. a pension fund is attracted to long-dated, index-linked bonds to match long-term, inflation-linked liabilities,
- or it can be due to regulatory reasons, taxation benefits, etc, e.g. an investor not liable to income tax would be more interested in high-coupon bonds.
- The yield for each "segment" of the market are determined by supply and demand in that segment, e.g. yield on short-term dated bonds is determined by supply and demand among short-term investors.)

For example, the Market Segmentation Theory can explain why yields at the long-end of the yield curve may decrease. However, for the overall shape of the yield curve, we need the Expectations Theory and Liquidity Preference Theory.

time	0	1	2	3	4
Random interest rate		i_1	i_2	i_3	i_4
$\overline{2}$| -ZCB	$-P_2$		100		
$\overline{4}$| -ZCB	$-P_4$				100
FI bond	$-P$	C	C	C	$C+R$

9.2 Zero-Coupon Bonds (ZCBs) and Spot Rates

Definition 9.1

£1 nominal of a n-year **zero-coupon bond (ZCB)** pays £1 after n years and nothing else, i.e. a ZCB is a fixed-interest security which is redeemed at par, with term n years and has **no** coupon payments.

In the UK market, a ZCB is sometimes called a "gilt strip", a gilt which has been stripped of its interest payments, or indeed just one of the coupon payments.

Definition 9.2

The **n-year spot rate of interest** y_n is the annual redemption yield of an n-year ZCB.

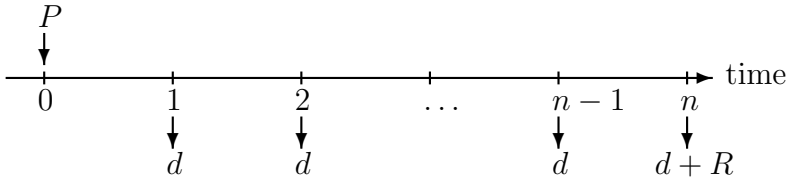
Let P_n be the price today of an n-year ZCB. Then, for $n > 0$,

$$\text{EoV:} \qquad P_n = 1 \times (1+y_n)^{-n} \Rightarrow y_n = P_n^{-\frac{1}{n}} - 1$$

Example 9.1 Calculate the spot rate of interest y_5 for a 5-year ZCB currently priced at 0.78.

Ans: $y_5 = 0.78^{\frac{1}{5}} - 1 = 5.095\%$ p.a.

Consider a fixed-interest security which pays redemption proceeds of R at time n, and dividends (coupons) d at times $1, 2, \ldots, n-1, n$. With no-arbitrage, the price P of this security must be equal to the price of a series of ZCBs.



$$\begin{aligned} P &= d(P_1 + P_2 + \cdots + P_n) + RP_n \\ &= d((1+y_1)^{-1} + (1+y_2)^{-2} + \ldots + (1+y_n)^{-n}) + R(1+y_n)^{-n} \end{aligned}$$

If this were not true then we could make money by buying a series of ZCBs and selling the fixed-interest security (or vice versa) to make a profit ($\Rightarrow$ arbitrage). This relationship allows us to

consider only ZCBs.

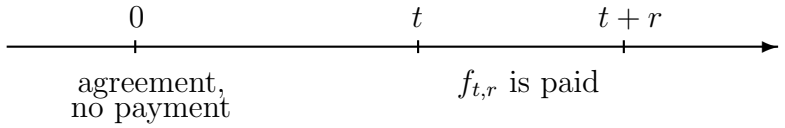
Exercise 9.2 Suppose $P_1 = 0.95$ and $P_2 = 0.90$. Calculate

1. the 1-year spot rate.
2. the 2-year spot rate.
3. the price of £100 nominal of a 2-year fixed-interest bond with coupon rate 7% p.a. paid annually in arrears and redemption proceeds 100%.
4. the gross redemption yield on the above fixed-interest bond.

9.3 Forward Rates

Definition 9.3

The *f*orward rate $f_{t,r}$ is the effective annual rate of interest agreed now (i.e. at time 0) for an investment made at time t for a period of r years.



What this means is that the present value **at time** t of a unit amount paid at time $t + r$ is

$$\frac{1}{(1 + f_{t,r})^r}.$$

The relationship between spot rates and forward rates

Exercise 9.3 Express $f_{0,r}$ in terms of a spot rate.

Consider £1 invested at time 0 for $t + r$ years at the $(t + r)$-spot rate. It will accumulate to $(1 + y_{t+r})^{t+r}$.

Alternatively, if we invest the £1 for t years at the t-year spot rate y_t followed by investment from time t to $t + r$ at the forward rate $f_{t,r}$, £1 accumulates to

$$(1 + y_t)^t (1 + f_{t,r})^r.$$

Under no-arbitrage, these accumulations must be equal, i.e.

$$(1 + y_t)^t (1 + f_{t,r})^r = (1 + y_{t+r})^{t+r}.$$

Rearranging, we get

$$(1 + f_{t,r})^r = \frac{(1 + y_{t+r})^{t+r}}{(1 + y_t)^t}.$$

Definition 9.4

The *n-year par yield* is the coupon rate c_n payable annually in arrears per £1 nominal which would give an n-year bond with coupon rate c_n a price of £1 assuming the bond is redeemed at par.

That is, for a given zero-coupon yield curve, the n-year par yield c_n is given by

$$\begin{aligned} 1 &= c_n[(1+y_1)^{-1} + (1+y_2)^{-2} + \ldots + (1+y_n)^{-n}] + (1+y_n)^{-n} \\ &= c_n[P_1 + P_2 + \ldots + P_n] + P_n \end{aligned}$$

Exercise 9.4 Show that, in terms of zero-coupon bond prices, we have

$$c_n = \frac{1 - P_n}{\sum_{k=1}^{n} P_k}.$$

Example 9.5 Consider the following spot interest rates

Term (in years)	Spot rate (in % p.a.)
1	5
2	5.5
3	6

The 3-year par yield c_3 satisfies

$$1 = c_3(1.05^{-1} + 1.055^{-2} + 1.06^{-3}) + 1.06^{-3} \qquad \Rightarrow \qquad c_3 = 5.96\% \text{ p.a.}$$

9.4 Continuous-time rates

Definition 9.5

The *t-year spot force of interest* is the force of interest equivalent to the annual spot rate y_t, i.e.

$$Y_t = \ln(1 + y_t).$$

Note that $P_t = e^{-tY_t}$. Y_t is also called

- the continuously-compounded spot rate, or
- the continuous-time spot rate.

Example 9.6 $y_t = 5\% \Rightarrow Y_t = 0.04879 = 4.879\%$.

Definition 9.6

The *continuous-time forward rate* $F_{t,r}$ is the force of interest equivalent to the annual rate of interest $f_{t,r}$, i.e.

$$F_{t,r} = \ln(1 + f_{t,r}).$$